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ENERGY CONTROL SYSTEM

Abstract

The present disclosure provides a method for controlling an energy control system. The method includes detecting a power outage at a grid interconnection. The method includes connecting a first set of the plurality of backup loads to a backup load interconnection, in which the backup load interconnection is electrically coupled to a backup power interconnection. The method includes disconnecting a second set of the plurality of backup loads from the backup load connection such that power is interrupted between the backup power source and the second set of backup loads. The method includes receiving a request from a user device to connect at least one load of the second set of the plurality of backup loads to the backup interconnection. The method includes determining whether to connect one load of the second set of the plurality of backup loads to the backup interconnection according to one or more programmed rules.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND INCORPORATED BY REFERENCE [0001] The present application is a continuation of U.S. patent application Ser. No. 18/394,873, filed Dec. 22, 2023, which is a continuation of U.S. patent application Ser. No. 17/520,277, filed Nov. 5, 2021, which is a continuation of U.S. application Ser. No. 16/811,832, filed Mar. 6, 2020, which claims priority to U.S. Provisional Patent Application No. 62/836,494, filed on Apr. 19, 2019, U.S. Provisional Patent Application No. 62/884,808, filed Aug. 9, 2019, and U.S. Provisional Patent Application No. 62/903,526, filed Sep. 20, 2019, each of which are incorporated by reference herein and made part of this specification.

FIELD
[0002] The present disclosure relates to energy control systems. In particular, embodiments relate to energy control systems for solar electric systems, for example, solar electric systems for residential buildings.
BACKGROUND
[0003] Some existing energy control systems may only offer partial home backup and may require the user to select loads to wire to a separate load panel that are powered through the inverter at all times. Other systems provide whole home and partial home backup through a central disconnect, but require multiple subpanel boxes and have limited controls. Furthermore, existing systems do not offer flexibility in energy system sizing and load management. Therefore, there is a need for an energy control system that offers the advantages of the features and functionalities of the present disclosure.
BRIEF SUMMARY
[0004] The present disclosure provides energy control systems for whole home and partial home backup with integrated breaker spaces and metering. In order to provide both whole home and partial home backup and to provide more advanced control application functionality to customers with two types of service panels (meter integrated and meter separated), the systems described herein have unique hardware features and software functionalities. For example, in some embodiments, the system may provide a separated and metered generation (PV and Storage) panel and breaker spaces. In some embodiments, the system may provide a separated, metered, and controllable essential load panel and breaker spaces. In some embodiments, the system may provide a separated and controllable non-backup load panel and breaker spaces. In some

embodiments, the system may provide metered and controllable electrical vehicle (“EV”) charger integration. In some embodiments, the system may provide for docking of a standard utility electric meter, for example, on the enclosure. In some embodiments, the system may provide for two, three, four, or more electric meters which allow for electric metering at various locations in the system. In some embodiments, the system may provide a central disconnect for a building (e.g., a house, apartment building, condominium, or multi-dwelling unit), to serve both whole home and partial home backup. In some embodiments, the system may provide for direct PV Smart Controls for optimal AC storage pairing in backup mode. Advantages of these features include, but are not limited to, advanced control options, reduced need for separate subpanels (e.g., for generation, essential loads, and large non-backup loads), increased flexibility and simplicity during installation, reduced installation time and cost, a more aesthetically pleasing design, and reduced cost to the end-user.

[0005] In some embodiments, an energy control system includes a microgrid interconnection device, a non-backup load interconnection, a backup load interconnection, and a photovoltaic interconnection. In some embodiments, the non-backup load interconnection, the backup load interconnection, and the photovoltaic interconnection may be electrically coupled to the microgrid interconnection device. In some embodiments, the energy control system may include a remotely-controllable electrical switch, and the remotely-controllable electrical switch may be electrically coupled to the backup load interconnection. In some embodiments, the energy control system may include an electric vehicle charger system, and the electric vehicle charger system may be electrically coupled to the backup load interconnection.

[0006] In some embodiments, an energy control system includes a grid interconnection electrically coupled to a utility grid, a backup power interconnection electrically coupled to a backup power source, a backup load interconnection electrically coupled to at least one backup load, and a non-backup load interconnection electrically coupled to at least one non-backup load. In some embodiments, the energy control system includes a microgrid interconnection device electrically coupled to the grid interconnection, the storage interconnection, the backup load interconnection, and the non-backup load interconnection. In some embodiments, the energy control system includes a controller in communication with the microgrid interconnection device. In some embodiments, the microgrid interconnection device is configured to switch between: (1) an on-grid mode electrically connecting the grid interconnection and the backup power interconnection to the backup and non-backup load interconnections, and (2) a backup mode electrically disconnecting the grid interconnection and the non-backup load interconnection from the storage interconnection.

[0007] In some embodiments, the controller is configured to detect a power outage at the grid interconnection, and upon detecting the power outage, the controller is configured to switch the microgrid interconnection device from the on-grid mode to the backup mode. In some embodiments, the controller is configured to detect a voltage restoration at the grid interconnection, and upon detecting the voltage restoration, the controller is configured to switch the microgrid interconnection device from the backup mode to the on-grid mode.

[0008] In some embodiments, the backup load interconnection includes a plurality of remotely-controllable switches electrically coupled to a plurality of backup loads. In some embodiments, each of the remotely-controllable switches is configured to switch between: (1) a closed position electrically connecting a respective backup load to the microgrid interconnection device, and (2) an open position electrically disconnecting the respective backup load from the microgrid interconnection device. In some embodiments, the controller is in communication with the plurality of remotely-controllable switches, and upon detecting the power outage at the grid interconnection, the controller is configured to switch one or more of the plurality of remotely-controllable switches from the closed position to the open position. In some embodiments, the controller switches the one or more remotely-controllable switches according to one or more programmed rules. In some embodiments, the one or more programmed rules are based on electronic data indicating energy and/or power consumption by the respective backup loads. In some embodiments, the controller is in communication with a user device, and upon receiving an input from the user device, the controller is configured to switch one or more remotely-controllable switches from the open position to the closed position or from the closed position to the open position.

[0009] In some embodiments, the backup load interconnection includes an electric vehicle charging port configured to charge electrical energy through the backup load interconnection to an electric vehicle and discharge electrical energy from the electric vehicle to the backup load interconnection. In some embodiments, the controller is in communication with the electric vehicle charging port, and upon detecting a power outage at the grid interconnection, the controller is configured to actuate the electric vehicle charging port to discharge electrical energy from the electric vehicle to the backup load interconnection.

[0010] In some embodiments, the backup power source includes a photovoltaic (PV) power generation system, and the backup power interconnection is configured to receive electrical energy generated by the PV power generation system. In some embodiments, the backup power source includes an energy storage system, and the backup power interconnection is configured to receive electrical energy discharged from the energy storage system.

[0011] In some embodiments, the energy control system includes an enclosure housing the grid interconnection, the backup power interconnection, the backup load interconnection, the non-backup load interconnection, the microgrid interconnection device, and the controller. In some embodiments, the energy control system includes an overcurrent protection device electrically disposed between the grid interconnection and the microgrid interconnection device.

[0012] In some embodiments, an electrical system includes a power generation system configured to generate electrical energy and an energy storage system configured to store electrical energy generated by the power generation system. In some embodiments, the electrical system includes an energy control system in communication with the power generation system and the energy storage system. In some embodiments, the energy control system includes a controller configured to receive electronic data from the energy storage system and configured to transmit a command to the power generation system to adjust a power output of the power generation system based on the electronic data from the energy storage system.

[0013] In some embodiments, the electronic data from the energy storage system indicates an available storage capacity of the energy storage system. In some embodiments, the available storage capacity corresponds to a difference between a total storage capacity and a current state of charge of the energy storage system. In some embodiments, the command transmitted by the controller indicates a desired power output of the power generation system corresponding to the available storage capacity of the energy storage system.

[0014] In some embodiments, the power generation system includes at least one power generation array having a photovoltaic (PV) panel array including one or more PV panels configured to generate electrical energy and a converter configured to adjust a power output of the PV panel array and the one or more PV panels. In some embodiments, the converter is configured to receive the command transmitted by the controller, and upon receiving the command, adjust the power output of the PV panel array such that the power output of the power generation system is adjusted according to the command. In some embodiments, the converter is configured to adjust the power output of the PV panel array by selectively enabling or disabling a corresponding PV panel array. In some embodiments, the converter is configured to adjust the power output of the PV panel array by adjusting a frequency of a power output of the PV panel array. In some embodiments, the controller determines a predicted power output of the power generation system based on a forecasted PV generation by the plurality of power generation arrays.

[0015] In some embodiments, the energy control system includes a load interconnection electrically coupled to a plurality of loads configured to consume electrical energy. In some embodiments, the energy control system is configured to receive electrical energy from the power generation system and the energy storage system. In some embodiments, the energy control system is configured to distribute the received electrical energy to the plurality of loads through the load interconnection. In some embodiments, the controller is configured to receive electronic data indicating an energy and/or power demand associated with the plurality of loads. In some embodiments, the energy demand is based on recorded data from detected energy consumption by at least one load of the plurality of loads. In some embodiments, the command transmitted by the controller indicates a desired power output of the power generation system corresponding to a sum of the energy demand associated with the plurality of loads and the available storage capacity of the energy storage system.

[0016] In some embodiments, an electrical system includes an energy control system. In some embodiments, the electrical system includes a photovoltaic (PV) power generation system electrically connected to the energy control system, and the PV power generation system is configured to generate electrical energy. In some embodiments, the electrical system includes an energy storage system electrically connected to the energy control system. In some embodiments, the energy storage system includes one or more energy storage units that are all interconnected on an AC terminal electrically coupled to the energy control system. In some embodiments, each storage unit includes a battery (e.g., a group of batteries) configured to be charged by the electrical energy generated by the power generation system and configured to discharge electrical energy to the energy control system. In some embodiments, each storage unit includes a storage converter (e.g., a bi-directional converter) configured to adjust a charging rate and a discharging rate of the battery. In some embodiments, the energy control system includes a controller configured to receive electronic data from the PV power generation system, the energy storage system, or both. In some embodiments, the controller is configured to transmit a command to the storage converter of at least one energy storage unit to adjust the charging rate or discharging rate of the battery.

[0017] In some embodiments, the command transmitted by the controller sets each battery of the one or more energy storage units at an equal charging rate or discharging rate.

[0018] In some embodiments, the electronic data indicates a predicted amount of electrical energy generated by the PV power generation system. In some embodiments, the command transmitted by the controller adjusts the charging rate or discharging rate of the battery based on the predicted amount of electrical energy generated by the PV power generation system. In some embodiments, the predicted amount of electrical energy generated by the PV power generation system is based on a weather forecast. In some embodiments, the predicted amount of electrical energy generated by the PV power generation system is based on prior generation information across a week or month or year.

[0019] In some embodiments, the electronic data indicates that a current state of charge of the battery has reached an upper threshold, and the command transmitted by the controller decreases the charging rate of the battery. In some embodiments, the upper threshold is in the range from approximately 90% to approximately 99% of a maximum battery storage capacity.

[0020] In some embodiments, the electronic data indicates that the current state of charge of the battery has reached a lower threshold, and the command transmitted by the controller decreases the discharging rate of the battery. In some embodiments, the lower threshold is in the range from approximately 1% to approximately 10% of a maximum battery storage capacity.

[0021] In some embodiments, the command transmitted by the controller indicates a state-of-charge mode, and upon receiving the command indicating the state-of-charge mode, the storage converter adjusts the charging rate or discharging rate of the battery to maintain the state of charge within a target state of charge. In some embodiments, the state-of-charge mode is a self-consumption mode, and the target state of charge in the self-consumption mode is in the range from approximately 1% to approximately 99% of a maximum battery storage capacity. In some embodiments, the state-of-charge mode is a backup mode, and the target state of charge in the backup mode is in the range from approximately 5% to approximately 95% of a maximum battery

storage capacity.
[0022] In some embodiments, the electrical control system includes a backup load interconnection electrically coupled to at least one backup load and a non-backup load interconnection electrically coupled to at least one non-backup load. In some embodiments, the controller is configured to receive load consumption data from the backup and non-backup load interconnections. In some embodiments, the controller is configured to transmit a command to the storage converter of at least one energy storage unit to adjust the charging rate or discharging rate of the battery based on the load consumption data.
[0023] In some embodiments, the load consumption data indicates a peak power consumption, and the command transmitted by the controller increases the discharging rate of the battery of at least one storage unit. In some embodiments, the load consumption data indicates an off-peak power consumption, and the command transmitted by the controller increases the charging rate of the battery of at least one storage unit.
[0024] In some embodiments, each of the energy storage units includes an enclosure housing the battery (e.g., a group of batteries) and the converter. In some embodiments, the enclosure includes a clamp configured to couple to the battery such that the battery is removably secured within the enclosure.
[0025] The present disclosure includes a method for controlling an energy control system having a grid interconnection, a backup load interconnection, a non-backup load interconnection, and a backup power interconnection. In some embodiments, the method includes a step of receiving electronic data from a plurality of backup loads. In some embodiments, the method includes a step of detecting a power outage at the grid interconnection electrically coupled to a utility grid. In some embodiments, the method includes a step of disconnecting the grid interconnection from the backup power interconnection, in which the backup power interconnection is electrically coupled to a backup power source. In some embodiments, the method includes a step of connecting a first set of the plurality of backup loads to the backup load interconnection, in which the backup load interconnection is electrically coupled to the backup power interconnection such that power is supplied from the backup power source to the first set of backup loads. In some embodiments, the first set of the plurality of backup loads is determined based on the electronic data from the plurality of backup loads.
[0026] In some embodiments, the method includes a step of disconnecting a second set of the plurality of backup loads from the backup load interconnection such that power is interrupted between the backup power source and the second set of backup loads.
[0027] In some embodiments, the second set of the plurality of backup loads is determined based on the electronic data from the plurality of backup loads.
[0028] In some embodiments, the electronic data indicates a detected power consumption and a usage time associated with each of the plurality of loads. In some embodiments, the electronic data includes data from a database defining a circuit load average associated with each of the plurality of loads with respect to discrete time blocks.
[0029] In some embodiments, the backup power source includes an electrical storage system. In some embodiments, the method includes a step of measuring a state of charge of the electrical storage system. In some embodiments, the first set of the plurality of backup loads is determined based on the measured state of charge of the electrical storage system. In some embodiments, a power load demand of the first set of the plurality of backup loads is less than the measured state of charge of the electrical storage system.
[0030] In some embodiments, the backup power source includes a photovoltaic (PV) power generation system. In some embodiments, the method includes a step of measuring a power output of the photovoltaic power generation system. In some embodiments, the first set of the plurality of backup loads is determined based on the measured power output of the photovoltaic power generation system. In some embodiments, a power load demand of the first set of the plurality of

backup loads is less than the measured power output of the photovoltaic power generation system.

[0031] The present disclosure includes methods for controlling an energy control system having a grid interconnection electrically connected to a utility grid, a backup load interconnection configured to be selectively electrically connected to a plurality of backup loads, and a backup power interconnection electrically connected to a backup power source. In some embodiments, the method includes a step of detecting a power outage at the grid interconnection. In some embodiments, the method includes a step of connecting a first set of the plurality of backup loads to the backup load interconnection, in which the backup load interconnection is electrically coupled to the backup power interconnection such that power is supplied from the backup power source to the first set of backup loads. In some embodiments, the method includes a step of disconnecting a second set of the plurality of backup loads from the backup load connection such that power is interrupted between the backup power source and the second set of backup loads. In some embodiments, the method includes a step of receiving a request from a user device to connect at least one load of the second set of the plurality of backup loads to the backup load interconnection. In some embodiments, the user device is a portable computing device. In some embodiments, the method includes a step of determining whether to connect the at least one load of the second set of the plurality of backup loads to the backup load interconnection according to one or more programmed rules.

[0032] In some embodiments, the method includes a step of receiving electronic data from the plurality of backup loads. In some embodiments, the first and second sets of the plurality of backup loads are determined based on the electronic data from the plurality of backup loads.

[0033] In some embodiments, the one or more programmed rules includes a step of determining an expected load demand associated with the at least one load of the second set of the plurality of backup loads and a step of comparing the expected load demand associated with the at least one load of the second set of the plurality of backup loads to an available power output of the backup power source. In some embodiments, the method includes a step of connecting the at least one load of the second set of the plurality of backup loads to the backup load interconnection based on the comparison between the expected load demand and the available power output of the backup power source. In some embodiments, the step of determining the expected load demand of the at least one load of the second set of the plurality of backup loads includes checking a database defining a circuit load average associated with each of the plurality of loads with respect to discrete time blocks.

[0034] In some embodiments, the backup power source is an electrical storage system, and the available power output is based on a measured state of charge of the electrical storage system. In some embodiments, the backup power source is a photovoltaic (PV) power generation system, and the available power output is based on a measured power output of the PV power generation system.

[0035] The present disclosure includes methods for controlling an electrical system having a power generation system, an energy storage system, and an energy control system, the energy control system electrically coupled to the power generation system, the energy storage system, and a plurality of loads. In some embodiments, the method includes a step of receiving electronic data from the power generation system, the energy storage system, the plurality of loads, or a combination thereof. In some embodiments, the method includes a step of determining a desired power output for the power generation system based on the received electronic data. In some embodiments, the method includes a step of determining an expected power output by the power generation system. In some embodiments, the method includes a step of comparing the expected power output to the desired power output to calculate a first adjustment to a power output of the power generation system. In some embodiments, the method includes a step of transmitting a first command indicating the first adjustment to the power generation system such that the power generation system adjusts the power output according to the first adjustment.

[0036] In some embodiments, the electronic data indicates a current state of charge and a total storage capacity of the energy storage system. In some embodiments, the desired power output corresponds to a difference between the total storage capacity and the current state of charge of the energy storage system.
[0037] In some embodiments, the electronic data indicates an energy demand associated with the plurality of loads. In some embodiments, the desired power output corresponds to the energy demand associated with the plurality of loads. In some embodiments, the energy demand is based on recorded data from detected energy consumption by the plurality of loads.
[0038] In some embodiments, the electronic data indicates an energy demand associated with the plurality of loads and a difference between a current state of charge and a total storage capacity of the energy storage system. In some embodiments, the desired power output corresponds to a sum of the energy demand associated with the plurality of loads and the difference between the current state of charge and the total storage capacity of the energy storage system.
[0039] In some embodiments, the power generation system includes a plurality of power generation arrays, each of the power generation arrays having a photovoltaic (PV) panel array configured to generate electrical energy and a converter configured to adjust a power output of the PV panel array. In some embodiments, the expected power output of the power generation system is determined based on a time of day, a weather forecast, a geographical location, a recorded power output by the power generation system, or any combination thereof. In some embodiments, the first command indicating the first adjustment is transmitted to at least one converter of the plurality of power generation arrays to enable or disable a corresponding PV panel array. In some embodiments, the first command indicating the first adjustment is transmitted to at least one converter of the plurality of power generation arrays to adjust a frequency of a power output of a corresponding PV panel array.
[0040] In some embodiments, the method includes a step of receiving power monitoring measurements indicating an actual power output of the power generation system. In some embodiments, the method includes a step of comparing the actual power output to the expected power output to calculate a second adjustment to the power output of the power generation system. In some embodiments, the method includes a step of transmitting a second command indicating the second adjustment to the power generation system such that the power generation system adjusts the power output according to the second adjustment.
[0041] In some embodiments, the second command indicates the second adjustment actuates the power generation system to increase the power output of the power generation system when the actual power output is lower than the estimated power output. In some embodiments, the second command indicates the second adjustment actuates the power generation system to decrease the power output of the power generation system when the actual power output is greater than the estimated power output.
[0042] In some embodiments, the method includes a step of exporting an excess power output generated by the power generation system through a utility interconnection of the energy control system, in which the utility interconnection is electrically coupled to a utility grid. In some embodiments, the excess power output corresponds to a difference between the actual power output and the expected power output.
[0043] In some embodiments, the method includes a step of exporting an excess power output generated by the power generation system to the energy storage system. In some embodiments, the excess power output corresponds to a difference between the actual power output and the expected power output.
[0044] In some embodiments, the method includes a step of exporting an excess power output generated by the power generation system to an electric vehicle battery through a backup load interconnection of the energy control system. In some embodiments, the excess power output corresponds to a difference between the actual power output and the expected power output.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0045] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate embodiments and, together with the description, further serve to explain the principles of the embodiments and to enable a person skilled in the relevant art(s) to make and use the embodiments.

[0046] FIG. **1** illustrates an electrical system according to an embodiment.

[0047] FIG. **2**A illustrates an electrical system according to an embodiment.

[0048] FIG. **2**B illustrates an electrical system according to an embodiment.

[0049] FIG. **3** illustrates an electrical system according to an embodiment.

[0050] FIG. **4**A illustrates an electrical system according to an embodiment.

[0051] FIG. **4**B illustrates an electrical system according to an embodiment.

[0052] FIG. **4**C illustrates an electrical system according to an embodiment.

[0053] FIG. **5**A illustrates an electrical system according to an embodiment.

[0054] FIG. **5**B illustrates an electrical system according to an embodiment.

[0055] FIG. **6** illustrates an electrical system according to an embodiment.

[0056] FIG. **7** illustrates an electrical system according to an embodiment.

[0057] FIGS. **8**A-B illustrate an electrical system according to an embodiment.

[0058] FIG. **9** illustrates a network according to an embodiment.

[0059] FIG. **10** illustrates a controller architecture according to an embodiment.

[0060] FIG. **11** illustrates an energy control system according to an embodiment.

[0061] FIG. **12** illustrates an energy control system according to an embodiment.

[0062] FIG. **13** illustrates a controller according to an embodiment.

[0063] FIG. **14** illustrates a microgrid interconnection device according to an embodiment.

[0064] FIG. **15** illustrates a block diagram of a controller according to an embodiment.

[0065] FIG. **16** illustrates an energy control system according to an embodiment.

[0066] FIG. **17**A illustrates an energy control system according to an embodiment.

[0067] FIGS. **17**B-**17**C illustrate an energy control system according to an embodiment.

[0068] FIGS. **18**A-**18**C illustrate an energy control system according to an embodiment.

[0069] FIGS. **19**A-**19**B illustrate an energy control system according to an embodiment.

[0070] FIGS. **20**A-**20**C illustrate an energy storage system according to an embodiment.

[0071] FIGS. **20**D-**20**G illustrate an energy storage system according to an embodiment.

[0072] FIG. **21** illustrates an energy storage system according to an embodiment.

[0073] FIGS. **22**A-**22**D illustrate an energy storage system according to an embodiment.

[0074] FIG. **23** illustrates an electrical system and a network according to an embodiment.

[0075] FIG. **24** illustrates an exemplary block diagram showing aspects of a method of controlling an energy control system according to an embodiment.

[0076] FIG. **25** illustrates an electrical system according to an embodiment.

[0077] FIG. **26** illustrates an electrical system according to an embodiment.

[0078] FIG. **27**A-**27**C illustrate an electrical system according to an embodiment.

[0079] FIG. **27**D illustrates an electrical system according to an embodiment.

[0080] FIG. **28** illustrates a block diagram showing aspects of a method of controlling an electrical system according to an embodiment.

[0081] FIG. **29** illustrates a diagram showing aspects of an energy storage system according to an embodiment.

[0082] FIG. **30** illustrates a user interface for an electrical system according to an embodiment.

[0083] FIGS. **31**A-**31**B illustrate schematics of the state of charging of an energy storage system according to embodiments.

[0084] FIG. **32** illustrates a timing diagram showing aspects of an electrical system according to an embodiment.
[0085] FIG. **33** illustrates an energy control system according to an embodiment.
[0086] FIG. **34** illustrates a microgrid interconnection device with a load-side overcurrent protection device according to an embodiment.
[0087] FIG. **35** illustrates an electrical system according to an embodiment.
[0088] FIGS. **36**A-**36**C illustrate graphs showing charging profiles of an electrical system according to embodiments.
[0089] FIG. **37** illustrates a block diagram showing aspects of a method of controlling an energy control system according to an embodiment.
[0090] FIGS. **38**A-**38**B illustrate an energy control system according to an embodiment.
[0091] FIG. **39** illustrates an energy storage system according to an embodiment.
[0092] FIG. **40** illustrates an electrical system according to an embodiment.
[0093] The features and advantages of the embodiments will become more apparent from the detail description set forth below when taken in conjunction with the drawings. A person of ordinary skill in the art will recognize that the drawings may use different reference numbers for identical, functionally similar, and/or structurally similar elements, and that different reference numbers do not necessarily indicate distinct embodiments or elements. Likewise, a person of ordinary skill in the art will recognize that functionalities described with respect to one element are equally applicable to functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0094] Embodiments of the present disclosure are described in detail with reference to embodiments thereof as illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “some embodiments,” “certain embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.
[0095] The term “about” or “substantially” or “approximately” as used herein refer to a considerable degree or extent. When used in conjunction with, for example, an event, circumstance, characteristic, or property, the term “about” or “substantially” or “approximately” can indicate a value of a given quantity that varies within, for example, 1-15% of the value (e.g., ±1%, ±2%, ±5%, ±10%, or ±15% of the value), such as accounting for typical tolerance levels or variability of the embodiments described herein.
[0096] The following examples are illustrative, but not limiting, of the present embodiments. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in the field, and which would be apparent to those skilled in the art, are within the spirit and scope of the disclosure.
[0097] FIG. **1** shows an electrical system **100** according to some embodiments. In some embodiments, electrical system **100** may include an energy control system **110** (e.g., “HUB+”). In some embodiments, energy control system **110** may control the flow of energy between an energy storage system **150**, a photovoltaic (“PV”) system **160**, electrical grid power **180**, and an electrical load **170**. Energy control system **110** may include, for example, batteries **158** for storing energy and a converter (e.g., an inverter) **152** for converting direct current (“DC”) to alternating current (“AC”) or vice versa. In some embodiments, energy storage system **150** may be expandable, which is to say that additional energy storage (e.g., additional batteries **158**) may be added to the system after initial installation. In some embodiments, photovoltaic system **160** may include, for example, one or more photovoltaic panels. In some embodiments, a plurality of photovoltaic panels may be

disposed on a building, for example, a residential home, office building, storage facility, or the like. In some embodiments, photovoltaic system **160** may be an AC coupled system and may include micro-inverters. In some embodiments, photovoltaic system **160** may include one or more string inverters. In some embodiments, photovoltaic system **160** may include other devices and/or systems for converting direct current ("DC") to alternating current ("AC") or vice versa.

[0098] In some embodiments, energy control system **110** may include a grid interconnection **184** electrically coupled to grid power **180**. In the context of the present disclosure, an interconnection includes any suitable electrical structure, such as a power bus, wiring, a panel, etc., configured to establish electrical communication between two set of circuits. In some embodiments, electrical system **100** may include an electrical meter **182** (e.g., "M"). In some embodiments, an electrical meter **182** may be electrically disposed between energy control system **110** and grid power **180** and may be used to measure and/or record the amount of electrical energy passing through the meter. In some embodiments, electrical meter **182** may measure and/or record the amount of electrical energy passing through the meter from grid power **180** to energy control system **110** and/or vice versa. Electrical load **170** may be, for example, one or more devices or systems that consume electricity. In some embodiments, electrical load **170** may include all or some of the electrical devices associated with a building. In some embodiments, electrical load **170** may include 240 volt loads. In some embodiments, electrical load **170** may include, for example, an electric range/oven, an air conditioner, a heater, a hot water system, a swimming pool pump, and/or a well pump. In some embodiments, electrical load **170** may include 120 volt loads. In some embodiments, electrical load **170** may include, for example, power outlets, lighting, networking and automation systems, a refrigerator, a garbage disposal unit, a dishwasher, a washing machine, a septic pump, and/or an irrigation system. In some embodiments, electrical load **170** may be separated into backup load **172** (e.g., one or more essential loads) and non-backup load **174** (e.g., one or more non-essential loads). In some embodiments, when electrical system **100** does not receive electricity from grid power **180** (e.g., during a power outage), backup load **172** may continue to receive power from energy storage system **150** and/or photovoltaic system **160**, while non-backup load **174** does not receive power from energy storage system **150** or photovoltaic system **160**.

[0099] FIG. **2**A shows an electrical system **200** according to some embodiments. In some embodiments, energy control system **210** may be electrically disposed between grid power **280** and a main service panel **270**. Main service panel **270** may, for example, distribute power to backup load **272** and may include overcurrent protection devices. In some embodiments, energy control system **210** may include a photovoltaic supervisory or monitoring system **230** (e.g., "PVS6") that monitors, for example, the status and/or performance of photovoltaic system **260**. In some implementations, the photovoltaic supervisory or monitoring system **230** can further control or modify operation of the PV system and/or provide communication functions relating to PV operation. In some embodiments, photovoltaic supervisory or monitoring system **230** may include the same features of and operate in the same manner as the photovoltaic supervisor system described in U.S. patent application Ser. No. 14/810,423, filed Jul. 27, 2015, the disclosure of which is incorporated herein in its entirety by reference thereto.

[0100] In some embodiments, energy control system **210** may include what is referred to herein synonymously as a microgrid interconnection device and/or an automatic transfer and/or disconnect switch **222** (e.g., "MID" or "ATS" or "ADS"). Microgrid interconnection device as described herein may be, for example, any device or system that is configured to automatically connect circuits, disconnect circuits, and/or switch one or more loads between power sources. In some embodiments, microgrid interconnection device **222** may include any combination of switches, relays, and/or circuits to selectively connect and disconnect respective circuits electrically coupled to energy control system **210**. In some embodiments, such switches may be automatic disconnect switches that are configured to automatically connect circuits and/or disconnect circuits. In some embodiments, such switches may be transfer switches that are configured to automatically switch one or more loads between power sources. In some embodiments, electrical system **200** may include a main overcurrent protection device **212** (e.g., “Primary OCPD”) that is electrically disposed between grid power **280** and components of energy control system **210**. In some embodiments, an electrical meter **282** may be electrically disposed between grid power **280** and energy control system **210**. In some embodiments, energy control system **210** may receive power from photovoltaic system **260**. In some embodiments, energy control system **210** may control the flow of power to and/or from energy storage system **250**. In some embodiments, energy control system **210** includes one or more load meters **240** that monitor the flow of electricity between certain elements of the electrical system **200**. In some embodiments, energy control system **210** may include ports for charging an electric vehicle (e.g., “EV Charger”).

[0101] In some embodiments, energy control system **210** may include a controller **220** (e.g., “PCBA”) that controls various operations of energy control system **210**. For example, controller **220** may control, for example, microgrid interconnection device **222**, energy storage system **250**, photovoltaic system **260**, a generator, and/or electric vehicle charging. In some embodiments, controller **220** may include a power supply, a MID controller that controls microgrid interconnection device **222**, a communications module (e.g., for Ethernet, RS-485, controller area network (CAN), power line communication (PLC), Wi-Fi™, and/or cellular), a generator control that controls a generator (described in further detail below), and/or an external jump start connector. In some embodiments, energy control system **210** may communicate wirelessly (e.g., send and/or receive data) via a wireless communication module **232**. In some embodiments, electrical system **200** may include backup loads **272** and non-backup loads **274**. In some embodiments, backup loads **272** may receive power from energy storage system **250** when electrical system **200** does not receive electricity from grid power **280**, while non-backup loads **274** may not receive power from energy storage system **250** when electrical system **200** does not receive electricity from grid power **280**.

[0102] FIG. **2**B shows electrical system **200** in another configuration. Electrical system **200** as shown in FIG. **2**B may include features and/or functionality similar to or the same as electrical system **200** described above with reference to FIG. **2**A. With reference to FIG. **2**A, in some embodiments, photovoltaic system **260** may be configured to provide power to backup loads **272** and non-backup loads **274** when microgrid interconnection device **222** is closed (set in on-grid mode), but may provide power only to backup loads **272** when microgrid interconnection device **222** is opened (set in backup mode, e.g., during a grid power outage). With reference to FIG. **2**B, in some embodiments, portion **260**A of photovoltaic system **260** may be configured to provide power to backup loads **272** and non-backup loads **274** when microgrid interconnection device **222** is closed, but may provide power only to backup loads **272** when microgrid interconnection device **222** is opened (set in backup mode, e.g., during a grid power outage). Further, portion **260**B of photovoltaic system **260** may be configured to provide power to backup loads **272** and non-backup loads **274** when microgrid interconnection device **222** is closed (set in on-grid mode), but may not provide power to backup loads **272** when microgrid interconnection device **222** is opened (set in backup mode, e.g., during a grid power outage).

[0103] FIG. **3** shows an electrical system **300** according to some embodiments. In some embodiments, energy control system **310** may be electrically disposed between main service panel **370** and backup loads **372**. In some embodiments, electrical system **300** may include backup loads **372** and non-backup loads **374**. In some embodiments, backup loads **372** may receive power from energy storage system **350** when electrical system **300** does not receive electricity from grid power **380**, while non-backup loads **374** may not receive power from energy storage system **350** when electrical system **300** does not receive electricity from grid power **380**. In some embodiments, backup loads **372** may be connected directly to ports in energy control system **310**. In some embodiments, backup loads **372** may be connected to energy control system **310** via a sub panel **376**. In some embodiments, a generator may be interconnected to energy control system **310** via sub panel **376**. In some embodiments, sub panel **376** may include a main subfeed overcurrent protection device **377** that is electrically disposed between, for example, backup loads **372** and the source of power for sub panel **376**. In some embodiments, electrical system **300** may include a main service panel **370**. Main service panel **370** may, for example, distribute power to non-backup loads **374** and may include overcurrent protection devices. In some embodiments, main service panel **370** may be electrically interconnected to energy control system **310** and may include an overcurrent protection device that is electrically disposed between grid power **380** and energy control system **310**.

[0104] In some embodiments, energy control system **310** may include a photovoltaic monitoring system **330** that monitors, for example, the status and/or performance of photovoltaic system **360**. In some embodiments, photovoltaic monitoring system **330** may be electrically connected to and in communication with (directly or indirectly) energy storage system **350**. In some embodiments, energy control system **310** may include a microgrid interconnection device **322**. In some embodiments, main service panel **370** may be electrically disposed between grid power **380** and microgrid interconnection device **322**. In some embodiments, an electrical meter **382** may be electrically disposed between grid power **380** and main service panel **370**. In some embodiments, electrical system **300** may include a main overcurrent protection device **384** (e.g., “Primary OCPD”) that is electrically disposed between grid power **380** and main service panel **370**. In some embodiments, main overcurrent protection device **384** may be integrated into main service panel **370**. In some embodiments, energy control system **310** may receive power from photovoltaic system **360**. In some embodiments, energy control system **310** may control the flow of power to and/or from energy storage system **350**. In some embodiments, energy control system **310** includes one or more load meters **340** that monitor the flow of electricity between certain elements of the electrical system **300**. In some embodiments, main service panel **370** includes a load meter **340** that monitors the flow of electricity through main service panel **370**. In some embodiments, energy control system **310** may include ports for charging an electric vehicle.

[0105] In some embodiments, energy control system **310** may include a controller **320** that controls various operations of energy control system **310**. For example, controller **320** may control, for example, microgrid interconnection device **322**, energy storage system **350**, photovoltaic system **360**, and/or electric vehicle charging. In some embodiments, controller **320** may include a power supply, a MID controller that controls microgrid interconnection device **322**, a communications module (e.g., for Ethernet, RS-485, Wi-Fi, and/or cellular), a generator control that controls a generator (described in further detail below), and/or an external jump start connector. In some embodiments, energy control system **310** may communicate wirelessly (e.g., send and/or receive data) via a wireless communication module **332**.

[0106] FIGS. **4**A-**4**C illustrate electrical system **400** according to some embodiments. As shown in FIGS. **4**A-**4**C, in some embodiments, electrical system **400** may include a generator **490**. Generator **490** may be, for example, powered by a fuel such as natural gas, propane, gasoline, oil, or the like. Generator **490** may be, for example, a portable generator (e.g., gas-powered), a standby generator, a home-backup generator, a smart generator, or the like. In some embodiments, electrical system **400** may include energy control system **410**, which may receive power from photovoltaic system **460** and/or electrical grid **480**. In some embodiments, electrical system **400** may include energy storage system **450**. In some embodiments, energy control system **410** may control the flow of power to and/or from energy storage system **450**. In some embodiments, electrical system **400** may include a microgrid interconnection device **492** that controls the flow of electricity between generator **490** and other components of electrical system **400**. In some embodiments, electrical system **400** may include a main service panel **470** that distributes power to electrical loads **472**.

[0107] FIG. **5**A shows an electrical system **500**, which may be used to control power from AC and DC, according to some embodiments. In some embodiments, electrical system **500** includes an energy control system **510**. In some embodiments, energy control system **510** includes a controller

520. In some embodiments, energy control system **510** includes a microgrid interconnection device **522**. In some embodiments, energy control system **510** includes a photovoltaic monitoring system **530**. In some embodiments, electrical system **500** includes an energy storage system **550**. In some embodiments, energy storage system **550** includes one or more converters **552**. In some embodiments, energy storage system **550** includes one or more batteries **558**. In some embodiments, electrical system **500** may be configured to allow an automatic “black start.” In some embodiments, electrical system **500** may be configured to allow a manual jumpstart (e.g., a “black start,” “dark start,” or the like) using, for example, jumpstart terminals **524**.

[0108] FIG. 5B shows another electrical system **500** according to some embodiments. In some embodiments, electrical system **500** may be configured to permit current flow in several directions (e.g., bidirectional) simultaneously between certain portions of electrical system **500**. For example, in some embodiments, electrical system **500** may include parallel conductors **555** (e.g., electrical wires) that allow current to flow both from energy storage system **550** and to energy storage system **550** simultaneously. For example, in some embodiments, electrical system **500** may include parallel conductors **555** that allow current to flow between energy storage system **550** and controller **520** simultaneously. In some embodiments, the power flowing between energy storage system **550** and controller **520** is DC power (e.g., 12V DC).

[0109] In some embodiments, parallel conductors **555** may allow current to simultaneously flow to, from, and between certain portions of energy storage system **550**. For example, in some embodiments, parallel conductors **555** may allow current to simultaneously flow to, from, and/or between one or more converters **552** of energy storage system **550**. Likewise, in some embodiments, parallel conductors **555** may allow current to simultaneously flow to, from, and/or between one or more batteries **558** of energy storage system **550**. In some embodiments, parallel conductors **555** may include one or more flow control devices **554** that each conducts current primarily in one direction. Flow control devices **554** may be, for example, diodes, semiconductor diodes, thermionic diodes, vacuum tubes, or other devices with asymmetric conductance. In this manner, flow control devices **554** may limit the flow of current through parallel conductors **555** to one direction.

[0110] In some embodiments, current may flow from controller **520** to converters **552**, and also between converters **552**, simultaneously. Similarly, current may flow from controller **520** to batteries **558**, and also between batteries **558**, simultaneously. During a system startup, for example, power from energy storage system **550** may be used to initiate controller **520** and/or inverter **526**. Since power may also flow from controller **520** to energy control system **550**, however, power from grid **580**, power generation system **560**, and/or a jumpstart battery (e.g., via jumpstart terminals **524** of FIG. 5A) may also be used to initiate converters **552**. In some embodiments, power shared between certain portions of electrical system **500** may be used to initiate other portions of energy control system **550** (e.g., photovoltaic monitoring system **530**).

[0111] In some embodiments, controller **520** may be configured to communicate with energy storage system **550**. For example, in some embodiments, controller **520** may be configured to control the state of energy storage system **550** (e.g., whether or not the system provides power to other portions of electrical system **500**) or the state of portions of energy control system **550** (e.g., particular converters **552** and/or batteries **558**). In some embodiments, converters **552** and/or batteries **558** may receive commands from controller **520** and may change their state (e.g., whether or not the converters and/or batteries provide power to other portion of electrical system **500**) based on the commands received from controller **520**. In some embodiments, batteries **558** may receive commands from converters **552** and may change their state (e.g., whether or not the batteries provide power to other portion of electrical system **500**) based on the commands received from converters **552**. In some embodiments, energy storage system **550** may include one or more status indicators **556** configured to communicate (e.g., to a homeowner or service technician) the state of energy storage system **550**. In some embodiments, status indicators **556** may be lights (e.g., light emitting diodes) that are illuminated when portions of energy storage system **550** are energized.

[0112] FIG. **6** shows an electrical system **600** according to some embodiments. In some embodiments, electrical system **600** includes an energy control system **610**. In some embodiments, energy control system **610** may include a grid interconnection **684** electrically coupled to a grid power **680**. In some embodiments, energy control system **610** includes an overcurrent protection device **612** between grid power **680** and other components of energy control system **610**. In some embodiments, energy control system **610** includes a non-backup load interconnection **614**, a backup load interconnection **616**, and/or a photovoltaic interconnection **618**. In some embodiments, non-backup load interconnection **614** may be, for example, a non-backup load interconnection panel, service panel, sub-panel, or the like. In some embodiments, backup load interconnection **616** may be, for example, a backup load interconnection panel, service panel, sub-panel, or the like. In some embodiments, non-backup loads may be connected to energy control system **610** using non-backup load interconnection **614**. In some embodiments, backup loads may be connected to energy control system **610** using backup load interconnection **616**. In some embodiments, photovoltaic system **660** may be connected to energy control system **610** using photovoltaic interconnection **618**. In some embodiments, energy control system **610** includes an electric vehicle connection **619**. In some embodiments, an electrical vehicle may be connected to and charged using electric vehicle connection **619**.

[0113] In some embodiments, energy control system **610** includes a controller **620**. In some embodiments, energy control system **610** includes a microgrid interconnection device **622**. In some embodiments, energy control system **610** includes a rapid shutdown switch **624**. In some embodiments, rapid shutdown switch **624** may be an emergency shutdown switch. As described in further detail below, in some embodiments, rapid shutdown switch **624** may be configured to, for example, de-energize portions of photovoltaic system **660**, energy storage system **650**, and/or other portions of electrical system **600**. In some embodiments, energy control system **610** includes a photovoltaic monitoring system **630**. In some embodiments, photovoltaic monitoring system **630** includes one or more antennas **632** for sending and/or receiving data over a wireless network. In some embodiments, energy control system **610** includes one or more load meters **640** that monitor the flow of electricity through certain elements of electrical system **600**. For example, a load meter **640** may monitor the flow of electricity from microgrid interconnection device **622** to backup load interconnection **616**. A load meter **640** may monitor the flow of electricity from microgrid interconnection device **622** to photovoltaic interconnection **618**. A load meter **640** may monitor the flow of electricity from grid power **680** to microgrid interconnection device **622**. Other locations for load meters **640** are also contemplated.

[0114] In some embodiments, electrical system **600** includes an energy storage system **650**. In some embodiments, energy storage system **650** includes one or more converters **652**. In some embodiments, energy storage system **650** includes one or more batteries **658**. In some embodiments, electrical system **600** includes a photovoltaic system **660**. In some embodiments, electrical system includes a main service panel **670**. In some embodiments, electrical system **600** includes a connection to grid power **680**. In some embodiments, an electrical meter **682** may be electrically disposed between energy control system **610** and grid power **680**.

[0115] FIG. **7** shows an electrical system **700** according to some embodiments. In some embodiments, electrical system **700** includes an energy control system **710**. In some embodiments, energy control system **710** includes a non-backup load interconnection **714**, a backup load interconnection **716**, and/or a photovoltaic interconnection **718**. In some embodiments, non-backup loads may be connected to energy control system **710** using non-backup load interconnection **714**. In some embodiments, backup loads may be connected to energy control system **710** using backup load interconnection **716**. In some embodiments, photovoltaic system **760** may be connected to energy control system **710** using photovoltaic interconnection **718**. In some embodiments, energy control system **710** includes an electric vehicle connection **719**. In some embodiments, an electrical vehicle may be connected to and charged using electric vehicle connection **719**.

[0116] In some embodiments, energy control system **710** includes a controller **720**. In some embodiments, energy control system **710** includes a microgrid interconnection device **722**. In some embodiments, energy control system **710** includes a rapid shutdown switch **724**. In some embodiments, energy control system **710** includes a photovoltaic monitoring system **730**. In some embodiments, photovoltaic monitoring system **730** includes one or more antennas **732** for sending and/or receiving data over a wireless network. In some embodiments, energy control system **710** includes one or more load meters **740** that monitor the flow of electricity through certain elements of electrical system **700**. For example, a load meter **740** may monitor the flow of electricity from microgrid interconnection device **722** to backup load interconnection **716**. A load meter **740** may monitor the flow of electricity from microgrid interconnection device **722** to photovoltaic interconnection **718**. A load meter **740** may monitor the flow of electricity from grid power **780** to main service panel **770**. Other locations for load meters **740** are also contemplated.

[0117] In some embodiments, electrical system **700** includes an energy storage system **750**. In some embodiments, energy storage system **750** includes one or more converters **752**. In some embodiments, converters **752** may be connected in parallel to one another. In some embodiments, energy storage system **750** includes one or more batteries **758**. In some embodiments, batteries **758** may be connected to converters **752** individually, serially, or in parallel. In some embodiments, electrical system **700** includes a photovoltaic system **760**. In some embodiments, electrical system includes a main service panel **770**. In some embodiments, electrical system **700** may include a connection to grid power **780**. In some embodiments, energy control system **710** may include a grid interconnection **784** forming part of the connection to grid power **780** such that the energy control system **710** is electrically coupled to grid power **784**. In some embodiments, an electrical meter **782** may be electrically disposed between energy control system **710** and grid power **780**, for example, between grid power **780** and main service panel **770**.

[0118] FIG. **33** shows an electrical system **3300** according to some embodiments. Electrical system **3300** may include the same or similar features as the embodiments the electrical systems described herein (e.g., electrical system **600**). In some embodiments, electrical system **3300** may include an energy control system **3310**, an energy storage system **3350** electrically coupled to energy control system **3310**, a PV system **3360** electrically coupled to energy control system **3310**, and a plurality of loads **3370** electrically coupled to energy control system **3310**. In some embodiments, the plurality of loads **3370** may be separated into a plurality of backup loads **3372** and a plurality of non-backup loads **3374**.

[0119] In some embodiments, energy control system may include a grid interconnection **3380** electrically coupled to a utility grid **3384**. In some embodiments, grid interconnection **3380** may include an AC bus **3381** and main service panel **3382** with a main circuit breaker **3383** (e.g., **200**A circuit breaker) electrically coupled to utility grid **3384** so that grid power is distributed to energy control system **3310**. In some embodiments, energy control system **3310** may include a non-backup load interconnection **3314** electrically coupled to the plurality of non-backup loads **3374**. In some embodiments, energy control system **3310** may include a power bus **3312** having a backup load interconnection **3316** electrically coupled to the plurality of backup loads **3372** and a storage interconnection **3317** electrically coupled to energy storage system **3350**. In some embodiments, energy control system **3310** may include a photovoltaic interconnection **3318** electrically coupled to the photovoltaic system **3360**. Any one of interconnections **3314**, **3316**, **3317**, and **3318** may include an AC bus, a panel, a sub-panel, a circuit breaker, or a combination thereof. In some embodiments, energy control system **3310** may include a controller **3320** configured to control the distribution of electrical energy between energy storage system **3350**, PV system **3360**, utility grid **3384**, and the plurality of loads **3370**.

[0120] In some embodiments, energy control system **3310** may include a microgrid interconnection device **3322** (e.g., an automatic transfer or disconnect switch) electrically coupled to grid interconnection **3380**, non-backup load interconnection **3314**, photovoltaic interconnection **3318**, and the combination of backup load interconnection **3316** and storage interconnection **3317** via power bus **3312**. In some embodiments, microgrid interconnection device **3322** may include any combination of switches, relays, and/or circuits to selectively connect and disconnect the grid interconnection **3380**, non-backup load interconnection **3314**, photovoltaic interconnection **3318**, and power bus **3312**. In some embodiments, controller **3320** is in communication with microgrid interconnection device **3322**.

[0121] As shown in FIG. **34**, for example, in some embodiments, microgrid interconnection device **3322** may include a first bus bar **3324** disposed at a line side of microgrid interconnection device **3322** and a second bus bar **3326** disposed at a load side of microgrid interconnection device **3322**. In some embodiments, first bus bar **3324** may be electrically coupled grid interconnection **3380** and non-backup load interconnection **3314**. In some embodiments, second bus bar **3326** may be electrically coupled to photovoltaic interconnection **3318** and power bus **3312** having backup load interconnection **3316** and storage interconnection **3317**. In some embodiments, as shown in FIG. **33** for example, energy control system **3310** may include a secondary overcurrent protection device **3328** disposed between the load side of microgrid interconnection device **3322** and the line side of power bus **3312**. In some embodiments, as shown in FIG. **34** for example, secondary overcurrent protection device **3328** may be a removable circuit breaker having a first end **3329** with a moveable contact **3329** disposed at second bus bar **3326** and a second end **3330** bolted to line side of power bus **3312**. In some embodiments, secondary overcurrent protection device **3328** is configured to interrupt power delivered to power bus **3312** when current at second bus bar **3326** reaches a maximum operating current.

[0122] In some embodiments, the maximum operating current associated with secondary overcurrent protection device **3328** and the current rating associated with second bus bar **3326** may be set to allow maximum distribution of PV power output while protecting the power bus **3312** from potentially harmful power surges. In some embodiments, the second bus bar **3326** is selected as a **400**A rated common bus bar, and secondary overcurrent protection device **3328** is set at a maximum operating current of 200 A. In some embodiments, power bus **3312** may include a **200**A rated bussing corresponding to secondary overcurrent protection device **3328** being rated at **200**A such that power bus **3312** may receive simultaneously **60**A from PV interconnection **3318**, **80**A from storage interconnection **3317**, and **60**A from grid interconnection **3380**.

[0123] FIG. **8**A shows an electrical system **800** according to some embodiments. In some embodiments, electrical system **800** includes an energy control system **810**, an energy storage system **850**, and/or a photovoltaic system **860**. In some embodiments, energy control system **810** may include a photovoltaic monitoring system **830**. In some embodiments, energy control system **810** may send and/or receive data (e.g., via photovoltaic monitoring system **830**) over a network **890**, for example, a wireless network. For example, energy control system **810** may send timeseries data and alerts from devices, receive tariff information (e.g., manually and/or automatically), receive control parameters and operational constraints, etc. In some embodiments, energy control system **810** may synchronize the transition from non-backup mode (where electrical system **800** is receiving electricity from grid power) to backup mode (where electrical system is not receiving electricity from grid power), using the microgrid interconnection device and energy storage system **850**. In some embodiments, photovoltaic monitoring system **830** may read timeseries data and/or disable a reconnection timer of photovoltaic system **860**. In some embodiments, photovoltaic monitoring system **830** may initiate a grid reconnection timer of photovoltaic system **860**. In some embodiments, photovoltaic monitoring system **830** may communicate with a battery monitoring system (“BMS”) of energy storage system **850**. In some embodiments, photovoltaic monitoring system **830** may communicate with energy storage system **850** and may, for example, read timeseries data, read power information, write charge/discharge targets, and/or write “heartbeats.” In some embodiments, photovoltaic monitoring system **830** may receive status and/or power information from a microgrid interconnection device.

[0124] In some embodiments, as shown in FIG. **8**A for example, energy control system **810** may include a controller **820** configured to monitor and adjust the status and operation of PV system **860** and energy storage system **850**. In some embodiments, controller **820** may be linked (e.g., wired or wirelessly) to PV monitoring system **830** such that controller **820** receives electronic data related to PV system **860** from PV monitoring system **830** and relays data received over network **890** to PV monitoring system **830**. In some embodiments, controller **820** may transmit commands to PV monitoring system **830** to adjust (e.g., increase or decrease) power output of PV system **860** based on data received over network **890** and electronic data received from PV monitoring system **830**. In some embodiments, controller **820** may be configured as a master controller and PV monitoring system **830** may be configured to communicate electronic data (e.g., status of power generation) with controller **820** such that controller **820** controls control energy distribution based on the electronic data transmitted by PV monitoring system **830**.

[0125] In some embodiments, as shown in FIG. **8**A for example, energy control system **810** may include a grid interconnection **880** electrically coupled to a utility grid. In some embodiments, grid interconnection **880** may include an AC bus **881**, main service panel **882**, and a utility meter **883** electrically coupled to a utility grid so that grid power is distributed to energy control system **810**. In some embodiments, main service panel **882** may be electrically coupled to a plurality of non-backup loads **874**. In some embodiments, energy control system **810** may include a backup power interconnection **840** electrically coupled to PV system **860** and energy storage system **850**. In some embodiments, backup power interconnection **840** may include any number of conductors (e.g., an AC power bus) to distribute electrical energy between energy control system **810**, energy storage system **850**, and PV system **860**. In some embodiments, energy control system **810** may include a backup load interconnection **870** electrically coupled to a plurality of backup loads **872** configured to consume electrical energy. In some embodiments, backup load interconnection **870** may include a load panel **871** having a plurality of switches to selectively distribute electrical energy to the plurality of backup loads **872**. In some embodiments, energy control system **810** may include a microgrid interconnection device **822** (e.g., a microgrid interconnection switch) electrically coupled to grid interconnection **880**, backup power interconnection **840**, and backup load interconnection **870**. In some embodiments, microgrid interconnection device **822** is configured to switch between: (1) an on-grid mode electrically connecting grid interconnection **880** and backup power interconnection **840** to backup load interconnection **870**, and (2) a backup mode electrically disconnecting grid interconnection **880** from the backup power interconnection **840** and backup load interconnection **870**. In some embodiments, PV system **860** may be coupled to main service panel **882** so that when microgrid interconnection device **822** is set in the backup mode, both PV system **860** and the utility grid are electrically disconnected from the backup power interconnection **840**. In some embodiments, energy storage system **850** may be coupled to main service panel **882** so that when microgrid interconnection device **822** is set in the backup mode, both energy storage system **850** and the utility grid are electrically disconnected from the backup power interconnection **840**.

[0126] In some embodiments, as shown in FIG. **8**A for example, energy control system **810** may include a housing **812** enclosing controller **820**, microgrid interconnection device **822**, PV monitoring system **830**, and backup power interconnection **870**, in which controller **820** is linked via hardwire to PV monitoring system **830**. In some embodiments, PV system **860**, energy storage system **850**, and energy control system **810** may each be disposed in a main residential building **802**. In some embodiments, as shown in FIG. **8**B for example, PV monitoring system **830** may be physically separated from housing **812** and linked wirelessly to controller **820** over a local-area-network **892** (e.g., residential Wi-Fi network). For example, in some embodiments, PV system **860** and PV monitoring system **830** may be disposed on an auxiliary building **804** (e.g., a garage) that is separate from the main residential building **802** (e.g., a house), where housing **812** having controller **820**, microgrid interconnection device **822**, and backup power interconnection **870** is located. In some embodiments, energy control system **810** may include a local-area-network access point (e.g., Wi-Fi access point) disposed in housing **812** so that controller **820** may communicate wirelessly with PV monitoring system **830** over the local-area-network **892**. In some embodiments, energy control system **810** may include a second access point (e.g., Bluetooth®) so that a user may employ a user device (e.g., portable computer device) to communicate directly with controller **820** via the second access point, for example, during a power outage that disables the local-area-network access point.

[0127] FIG. **9** shows a network **900** according to some embodiments. In some embodiments, an energy control system (e.g., any energy control system described herein) may send and or receive data via network **900**. In some embodiments, timeseries data, alerts, metadata, outage reports, power usage information, greenhouse gas generation reports, service codes, runtime data, etc. may be sent and/or received using network **900**. In some embodiments, the data may be stored locally (e.g., in the memory of controller **2330**, described below) for review, for example, by a homeowner and/or service personnel. In some embodiments, an administrator may remotely send commands using network **900**. In some embodiments, the information may be sent and/or received by a personal computing device, for example, a desktop computer, laptop computer, or mobile phone (e.g., using an application program).

[0128] FIG. **10** shows a controller architecture **1000** according to some embodiments. In some embodiments, information may be communicated along the pathways and between the system components as indicated in FIG. **10**. In some embodiments, controller architecture **100** may include a storage module **1010** adapted to collect information (e.g., operating status and electronic data) directly from converters of electronic storage system (e.g., electronic storage system **150**) and a photovoltaic (PV) supervisor module **1020** adapted to collect information (e.g., operating status and electronic data) directly from a PV system (e.g., PV system **160**). In some embodiments, storage module **1010** may include a storage interface **1012** configured to communicate with modules **1030** of electronic storage system. In some embodiments, storage module **1010** may include a PV interface **1014** (DBUS interface) configured to communicate with PV module **1020**. In some embodiments, storage module **1010** may include a processing sub-module **1016** configured to process information received from storage and PV interface sub-modules **1012**, **1014** and transmit commands to modules **1030** of the electronic storage system.

[0129] As shown in FIG. **11**, in some embodiments, energy control system **1100** may include a mounting plate **1110** to which components of energy control system **1100** may be mounted. As shown in FIG. **11**, in some embodiments, energy control system **1100** may include a controller **1120**, a microgrid interconnection device **1122**, and/or a photovoltaic monitoring system **1130**. In some embodiments, energy control system **1100** may include a main overcurrent protection device **1112**. As shown in FIG. **11**, in some embodiments, energy control system **1100** may include a non-backup load interconnection **1114**, a backup load interconnection **1116**, and/or a photovoltaic interconnection **1118**. In some embodiments, non-backup loads may be connected to energy control system **1100** using non-backup load interconnection **1114**. In some embodiments, backup loads may be connected to energy control system **1100** using backup load interconnection **1116**. In some embodiments, the photovoltaic system may be connected to energy control system **1100** using photovoltaic interconnection **1118**. Mounting plate **1110** may be sized and shaped to accommodate all of the components of energy control system **1100**. In some embodiments, mounting plate **1110** may be rectangular. In some embodiments, mounting plate **1110** may have a greater length than width. In some embodiments, mounting plate **1110** may be configured for indoor enclosures such as, for example, NEMA-1 enclosures. In some embodiments, mounting plate **1110** may be configured for outdoor and/or weather rated enclosures such as, for example, NEMA-3 enclosures. In some embodiments, the internal components of energy control system **1100** may be used with different types of enclosures without being reconfigured.

[0130] As shown in FIG. **12**, in some embodiments, energy control system **1200** may include a mounting plate **1210** on which components of energy control system **1200** may be mounted. As shown in FIG. **12**, in some embodiments, energy control system **1200** may include a controller **1220**, a microgrid interconnection device **1222**, and/or a photovoltaic monitoring system **1230**. In some embodiments, energy control system **1200** may include a main overcurrent protection device **1212**. As shown in FIG. **12**, in some embodiments, energy control system **1200** may include a photovoltaic interconnection **1218**. In some embodiments, energy control system **1200** may include load interconnections **1214**. In some embodiments, the photovoltaic system may be connected to energy control system **1200** using photovoltaic interconnection **1218**. In some embodiments, loads may be connected to energy control system **1100** using load interconnections **1214**. In some embodiments, electrical loads may be indirectly connected to energy control system **1200** using, for example, a main service panel and/or a sub panel.

[0131] As shown in FIG. **38**B for example, in some embodiments, energy control system **3800** may include a mounting plate **3810** on which components (e.g., controller, MID load pan) of energy control system **3800** may be mounted. In some embodiments, energy control system **3800** may include a housing **3820** receiving mounting plate **3810**. In some embodiments, energy control system may include a PV monitoring system **3830** or another monitoring/control system that is connected to housing **3820** or mounting plate **3810**, for example, by a hinge **3832** such that PV monitoring system **3830** is configured to pivot between an operating position and a service position. In some embodiments, as shown in FIG. **38**A for example, at the operating position, PV monitoring system **3830** may be disposed against or parallel with mounting plate **3810**. In some embodiments, as shown in FIG. **38**B for example, at the service position, PV monitoring system **3830** may be pivoted away from mounting plate **3810** to allow a user to access cabling or service other components secured to mounting plate **3810**.

[0132] FIG. **13** shows a controller **1300** (e.g., microgrid interconnection device controller PCBA) according to some embodiments. Controller **1300** may include any of the features disclosed herein related to controllers.

[0133] FIG. **14** shows a portion of a microgrid interconnection device **1400** according to some embodiments. Microgrid interconnection device **1400** may include any of the features disclosed herein related to microgrid interconnection devices.

[0134] FIG. **15** shows a block diagram of a controller **1500** (e.g., a microgrid interconnection device controller) according to some embodiments. In some embodiments, controller **1500** may include a microcontroller **1502** having a central-processing-unit (e.g., programmable system-on-chip controller). In some embodiments, controller **1500** may handle microgrid interconnection device coils of 12 volts using an ALT contactor drive **1504**. In some embodiments, controller **1500** may control relays. In some embodiments, controller **1500** may measure current and voltage (e.g. via current transformer “CT” or Potential transformer “PT”). In some embodiments, controller **1500** may measure the differences between grid power and backup power such as, for example, differences in phase, frequency, voltage, etc. In some embodiments, controller **1500** may have an internal power supply **1510** (e.g., 5 VDC & 3.3 VDC internal power supply). In some embodiments, controller **1500** may have external power inputs **1512** (e.g., 48V to 12 VDC power supply from storage battery). In some embodiments, controller **1500** may receive power from the electrical grid, an energy storage system, and/or a power generation system. In some embodiments, controller **1500** may control microgrid interconnection device coils that need to reverse coil polarity to reset. In some embodiments, controller **1500** may include an auxiliary power input to receive power from batteries. In some embodiments, controller **1500** may include connectors for communicating with and/or controlling the microgrid interconnection device. In some embodiments, controller **1500** may include a “test mode” for simulating a grid outage and/or an emergency system shutdown. In some embodiments, controller **1500** may include visual and/or audible alarms, for example, to indicate system status. In some embodiments, controller **1500** may allow a rapid system shutdown (including remote shutdown) and/or a direct pass-through to system inverters for safety. In some embodiments, controller **1500** may include generator start relays for continuous backup and/or battery jump-start relays for "dark-start" situations.

[0135] FIG. **16** shows an energy control system **1600** according to some embodiments. In some embodiments, energy control system **1600** includes a housing **1610**. In some embodiments, energy control system **1600** includes a cover **1620** for the photovoltaic monitoring system. In some embodiments, cover **1620** may be plastic. In some embodiments, cover **1620** may be removable. In some embodiments, energy control system **1600** includes a door (not shown) that may be opened and closed in order to access components mounted on mounting plate **1110** (see FIG. **11**), for example, overcurrent protection devices **1630**. In some embodiments, the door may be hinged. In some embodiments, energy control system **1600** may be used indoors. As described in further detail below, in some embodiments, energy control system **1600** may be, for example, mounted between the studs of a wall in a building. In some embodiments, energy control system **1600** with housing **1610** meets NEMA-1 enclosure standards. In some embodiments, housing **1610** may be about 12-16″ wide, about 40-50″ tall, and about 3-5″ deep. In some embodiments, housing **1610** may be about 13-15″ wide, about 43-47″ tall, and about 3.5-4.5″ deep. In some embodiments, housing **1610** may be about 14″ wide, about 45″ tall, and about 3.8″ deep. In some embodiments, housing **1610** may be about 20-40″ tall.

[0136] FIG. **17** shows an energy control system **1700** according to some embodiments. In some embodiments, energy control system **1700** includes a housing **1710**. In some embodiments, energy control system **1700** includes a cover **1720** for the photovoltaic monitoring system. In some embodiments, cover **1720** may be plastic, metal, or a combination of plastic and metal. In some embodiments, cover **1720** may be removable. In some embodiments, cover **1720** may be rotatably and/or removably connected to housing **1710**. In some embodiments, energy control system **1700** includes a door **1730** that may be opened and closed in order to access components mounted on mounting plate **1110** (see FIG. **11**). In some embodiments, the door may be hinged. In some embodiments, the door may include a latch for securing the door in a closed position. In some embodiments, the door may be lockable. In some embodiments, energy control system **1700** may be used outdoors. In some embodiments, housing **1710**, cover **1720**, and/or door **1730** may be sealed to prevent or reduce the amount of water, dirt, or debris, for example, from entering the unit. In some embodiments, a seal may be disposed between door **1730** and housing **1710**. As described in further detail below, in some embodiments, energy control system **1700** may be, for example, mounted to a wall of a building. In some embodiments, energy control system **1700** with housing **1710** meets NEMA-3 enclosure standards. In some embodiments, housing **1710** may include one or more openings and/or knockout holes disposed on several sides of housing **1710** for flexibility during installation of electrical conduit and/or wiring. In some embodiments, housing **1710** may be about 15-19″ wide, about 43-53″ tall, and about 4-6″ deep. In some embodiments, housing **1710** may be about 16-18″ wide, about 46-50″ tall, and about 4.5-5.5″ deep. In some embodiments, housing **1710** may be about 17″ wide, about 48″ tall, and about 5″ deep.

[0137] FIGS. **17**B-C show an energy control system **1700**B according to some embodiments. In some embodiments, energy control system **1700**B may include a housing **1710**B. In some embodiments, housing **1710**B may be plastic, metal, or a combination of plastic and metal. In some embodiments, energy control system **1700**B may include a cover **1720**B enclosing a photovoltaic monitoring system (e.g., PV monitoring system **1230**). In some embodiments, cover **1720**B may be plastic, metal, or a combination of plastic and metal. In some embodiments, cover **1720**B may be removable. In some embodiments, cover **1720**B may be rotatably and/or removably connected to housing **1710**B. In some embodiments, energy control system **1700**B may include a door **1730**B that is configured to be opened and closed to access components (e.g., switches) mounted within housing **1710**B, for example on a mounting plate (e.g., mounting plate **1110** shown in FIG. **11**). In some embodiments, door **1730**B may be hinged, for example having one or more hinges coupling door **1730**B and housing **1710**B. In some embodiments, door **1730**B may include one or more latch, for example an upper latch **1732**B and/or a lower latch **1732**C for securing door **1730**B in a closed position. The latches may increase the compression of the door **1730**B (e.g., against the seals or gaskets discussed below) to meet NEMA-3 enclosure standards. In some embodiments, door **1730**B may be magnetically coupled to housing **1710**B to secure door **1730**B in a closed position. In some embodiments, the door may be lockable. In some embodiments, door **1730**B may include a padlock, for example disposed along a bottom edge thereof to securely lock door **1730**B to housing **1710**B when set in the closed position.

[0138] In some embodiments, energy control system **1700**B may be used outdoors. In some embodiments, housing **1710**B meets NEMA-3 enclosure standards by forming a sealed enclosure with cover **1720**B and door **1730**B. In some embodiments, housing **1710**B and/or door **1730**B may include one or more gaskets **1712**B disposed along one or more side edges and an upper side thereof to form a seal between housing **1710**B and door **1370**B when set in a closed position. In some embodiments, housing **1710**B, cover **1720**B, and/or door **1730**B may include a gasket disposed along each longitudinal and upper side thereof to prevent or reduce the amount of water, dirt, or debris, for example, from entering the unit. In some embodiments, the gasket may be formed of, for example, a foam-based material having elastic properties.

[0139] FIGS. **18**A-**18**C show an energy control system **1800** according to some embodiments. In some embodiments, energy control system **1800** may include rear housing portion **1810** and a front housing portion **1820**. As shown in FIG. **18**B, in some embodiments, rear housing portion **1810** may be mounted between studs **1830** of a wall of a building. As shown in FIG. **18**C, in some embodiments, front housing portion **1820** may be mounted to a wall panel **1840** (e.g., drywall) of a wall of a building. Thus, energy control system **1800** and its associated housing(s) may be conveniently and securely mounted for indoor use. In some embodiments, for example, rear housing portion **1810** and mounting plate **1110** (see FIG. **11**) with associated energy control system components may be mounted between adjacent studs **1830**. A wall panel **1840** having an opening approximately the same size as rear housing portion **1810** may be attached to the studs **1830**. Front housing portion **1820** may be coupled to rear housing portion **1810**. In some embodiments, energy control system **1800** may be mounted flush with wall panel **1840**. In some embodiments, rear housing portion **1810** and front housing portion **1820** may be installed prior to other components of energy control system **1800**. For example, rear housing portion **1810** and front housing portion **1820** may be installed during initial construction of the building along with, for example, electrical conduit for routing electrical wiring. Then, a mounting plate with electrical components (e.g., mounting plate **1110**, described above) may be installed and wired after construction of the building is complete. Thus, homeowners may more easily integrate energy control system **1800** into their home even if it is not installed during initial construction, without requiring substantial alterations to the existing structure of the building.

[0140] FIGS. **19**A-**19**B show an energy control system **1900** according to some embodiments. Energy control system **1900** may have a housing and/or a door (see e.g., FIG. **17**, housing **1710** and door **1730**). As shown in FIG. **19**B, in some embodiments, energy control system **1900** may be mounted to a wall **1940** of a building.

[0141] FIGS. **20**A-**20**C show an energy storage system **2000** according to some embodiments. As shown in FIGS. **20**A and **20**C, in some embodiments, energy storage system **2000** may include a frame **2010**. As shown in FIG. **20**B, in some embodiments, energy storage system **2000** may include a housing **2020**. In some embodiments, housing **2020** may be ventilated. In some embodiments, housing **2020** may include an access door. In some embodiments, housing **2020** may include a dead front cover. In some embodiments, housing **2020** may include conduit openings, knockout holes, or the like for power and/or communication cables. In some embodiments, housing **2020** may include a window such that a user can see inside housing **2020** while the access door is closed. As shown in FIG. **20**C, in some embodiments, energy storage system **2000** may include a converter **2052** and one or more batteries **2058**. Converter **2052** and/or batteries **2058** may be coupled to, disposed on, and/or supported by frame **2010**. In some embodiments, frame **2010** includes slot(s) for batteries **2058**. In some embodiments, converter **2052** may be hung from frame **2010**. As shown in FIG. **20**C, in some embodiments, energy storage system **2000** may be disposed substantially in a vertical arrangement.

[0142] FIGS. **20**D-G show an energy storage system **2000**B according to some embodiments. In some embodiments, energy storage system **2000**B may include a housing **2020**B and a door **2030**B coupled to housing **2020**B, for example via a hinge **2025**B such that door **2030**B is configured to swing about the hinge between an open position and a closed position. In some embodiments, the hinge **2025**B may include half-barrels disposed along a side edge of door **2030**B and housing **2020**B, where the half-barrels of door **2030**B and housing **2020**B interlock to receive a pin. In some embodiments, housing **2020**B and/or door **2030**B may include latches disposed along a side edge of housing **2020**B or door **2030**B, and/or a padlock disposed, for example, along a bottom edge of housing **2020**B or door **2030**B to secure door **2030**B in the closed position. In some embodiments, door **2030**B may include a bottom section **2034** comprised of a metal-based material (e.g., aluminum, steel) and a top section **2032** comprised of a polymer-based material (e.g., polyethylene, polyethylene terephthalate, polyvinyl chloride, polypropylene, polycarbonate). In some embodiments, bottom section **2034** and top section **2032** of door **2030**B may be coupled to one another. In some embodiments, door **2030**B may include a display **2036** (e.g., user interface **2790** shown in FIG. **30**).

[0143] In some embodiments, housing **2020**B may be mounted to a wall and/or floor (e.g., the ground, a cement or tile floor, etc.) to secure energy storage system **2000**B in place and protect it from damage (e.g., from loose equipment stored near the energy storage system, seismic activity, etc.). In some embodiments, housing **2020**B may include one or more rails **2024**B and a perforated bracket **2026**B, for example disposed along the back of housing **2020**B, to mount housing **2020**B to a wall. In some embodiments, housing **2020**B may include one or more (e.g., two, three, or four) feet assembly **2040**B, for example disposed along a bottom of housing **2020**B. In some embodiments, feet assembly **2040**B may include a plurality of bolt-feet **2042**B projecting from the bottom of housing **2020**B and a floor mounting bracket **2044**B disposed at the end of feet **2042**B to mount to the floor. In some embodiments, feet assembly **2040**B may include concrete anchor bolts to secure bracket **2044**B to the floor. In some embodiments, the length of feet **2042**B may be adjustable such that feet assembly **2040**B is configured to provide a level base for housing **2020**B, even if mounted to an uneven floor surface. In some embodiments, feet assembly **2040**B is configured to comply with seismic compliance codes. In some embodiments, feet assembly **2040**B may be separately installed onto housing **2020**B.

[0144] In some embodiments, energy control system **2000**B may be used outdoors. In some embodiments, housing **2020**B meets NEMA-3 enclosure standards by forming a sealed enclosure with door **2030**B. In some embodiments, housing **2020**B and/or door **2030**B may include one or more gaskets **2022**B, for example disposed along the side edges and an upper side thereof to form a seal between housing **2020**B and door **2030**B when set in a closed position. In some embodiments, housing **2020**B and/or door **2030**B may include a gasket disposed along each longitudinal and upper side thereof to prevent or reduce the amount of water, dirt, or debris, for example, from entering the unit. In some embodiments, the gasket may be formed of a foam-based material having elastic properties.

[0145] In some embodiments, housing **2020**B is configured to receive one or more converter **2052**B and one or more batteries **2058**B. In some embodiments, converter **2052**B may be disposed within an upper portion of housing **2020**B, for example mounted to an interior wall of housing **2020**B. In some embodiments, housing **2020**B may include a battery mounting unit **2060**B, for example disposed within a lower portion of housing **2020**B and configured to hold one or more batteries **2058**B. In some embodiments, housing **2020**B may include a wire harness to hold wires extending between converter **2052**B and the one or more batteries **2058**B. In some embodiments, battery mounting unit **2060**B includes a bracket **2062**B, for example disposed on the bottom of housing **2020**B and configured to receive one or more batteries **2058**B. In some embodiments, battery mounting unit **2060**B includes a clamp **2064**B configured to couple to a top of the one or more batteries **2058**B. In some embodiments, clamp **2064**B is configured to snap over the one or more batteries **2058**B (e.g., over a top of a battery). In some embodiments, clamp **2064**B is configured to be pressure-fitted against the top of the batteries **2058**B, and the clamp **2064**B may include a knob **2066**B to adjust the tension of the pressure-fit against the one or more batteries **2058**B. By allowing adjustment of the pressure-fit or establishing securement by snapping, battery mounting unit **2060**B enables faster installation of the one or more batteries **2058**B within housing **2020**B. As such, to insert a battery, a user may lift clamp **2064**B, insert a battery onto battery mounting unit **2060**B (e.g., by sliding the battery onto battery mounting unit **2060**B such that a slot in the battery mates with the battery mounting unit **2060**B), and closing the clamp **2064**B to secure the battery in place. This arrangement increases the speed and ease with which the user can change the batteries.

[0146] In some embodiments, housing **2020**B and door **2030**B may be ventilated by forced air convection to maintain converter **2052**B and the one or more batteries **2058**B at an effective operating temperature. In some embodiments, housing **2020**B or door **2030**B may include an inlet vent **2072**, an outlet vent **2074**, and a fan **2070** to draw ambient air into housing **2020**B via inlet vent **2072** and propel heated air out of housing **2020**B via outlet vent **2074**. In some embodiments, outlet vent **2074** is disposed proximate to a top of housing **2020**B and inlet vent **2072** is disposed proximate to a bottom of housing **2020**B, such that ambient air is drawn through a bottom portion housing **2020**B and circulated to an upper portion of housing **2020**B. In some embodiments, inlet vent **2072** and outlet vent **2074** may include a set of louvers **2076** and/or an air filter to protect against ingress of solid foreign objects (e.g., dirt, dust) and the ingress of liquid (e.g., rain, sleet, snow) such that housing **2020**B complies with NEMA-3 standards. In some embodiments, the louvers of inlet vent **2072** and outlet vent **2074** are configured to protect against spraying liquids (e.g., jet-sprayed water) so that water is prevented from reaching the air filter of vents **2072**, **2074**.

[0147] FIG. **21** shows an energy storage system **2100** according to some embodiments. As shown in FIG. **21**, in some embodiments, energy storage system **2100** may include a frame **2110**, a converter **2152**, and batteries **2158**. As shown in FIG. **21**, in some embodiments, energy storage system **2100** may be disposed substantially in a horizontal arrangement. In some embodiments, energy storage system **2100** may be disposed indoors. In some embodiments, energy storage system **2100** may include cable raceways for power and/or communication cables.

[0148] FIG. **39** shows an energy storage system **3900** according to some embodiments. As shown in FIG. **39**, in some embodiments, energy storage system **3900** may include a converter **3952**, batteries **3958**, and an energy control system **3910**. In some embodiments, energy control system **3910** may be any one of the energy control systems (e.g., energy control system **810** shown in FIG. **8**A or energy control system **3310** shown in FIG. **33**). In some embodiments, energy storage system **3900** may include raceway conductors **3960** and **3962** electrically connecting batteries **3958** to converter **3952** and energy control system **3910**.

[0149] FIGS. **22**A-**22**D show an energy storage system **2200** according to some embodiments. As shown in FIGS. **22**A-**22**D, in some embodiments, energy storage system **2200** may include a frame **2210**. As shown in FIG. **22**C, in some embodiments, energy storage system **2200** may include a housing **2220**. In some embodiments, housing **2220** may include an access door. As shown in FIG. **22**B-**22**D, in some embodiments, energy storage system **2200** may include a converter **2252**. As shown in FIGS. **22**B and **22**D, in some embodiments, energy storage system **2200** may include batteries **2258**. In some embodiments, converter **2252** and/or batteries **2258** may be installed in a ventilated housing **2220**, for example, a housing that meets NEMA-3 enclosure standards. In some embodiments, energy storage system **2200** may be installed indoors or outdoors. In some embodiments, energy storage system **2200** may be modular and/or customizable based on the installation location.

[0150] Some smart devices (e.g., smart outlets, smart plugs, smart bulbs, or the like) may permit a homeowner to, for example, remotely control the state (e.g., on and off) of certain electrical devices (e.g., lamps and light fixtures) and/or monitor the power consumption of the electrical devices to which they are connected. Similarly, some smart appliances (e.g., smart washers, smart refrigerators, or the like) may be configured to be remotely controlled by a user and/or to monitor their power consumption. While such smart devices and smart appliances may afford a homeowner the ability to control their electrical devices remotely, they may be impractical and expensive to implement in a home, for example, if control over a large quantity of devices is desired.

[0151] For example, if a homeowner wishes to remotely control the lighting in several rooms of their home, it may be necessary to acquire and install a quantity of smart devices (e.g., smart outlets or smart bulbs) that may be proportional to the number lighting devices over which control is desired. Considering that a home may include many lights (as well as other electric devices that a homeowner may wish to remotely control) it may be expensive and impractical to purchase and install such a large quantity of smart devices in a home. Further, the homeowner may also need to configure and connect each individual device to a home network (e.g., a WIFI™ network), which may be cumbersome and time consuming. Accordingly, for such systems, as the quantity of controlled devices increases, the costs and complexity of installation may also increase.

[0152] Embodiments described herein may allow a user to remotely control and/or monitor the power consumption of many electrical devices in a practical and cost-efficient manner. Embodiments may include, for example, a remotely-controllable switch that interrupts the flow of current to an entire electrical circuit, thereby allowing one remotely-controllable switch to control the state of one or more electrical devices connected to the electrical circuit. Further, such a system may allow a homeowner to control and/or monitor the power consumption of “non-smart” appliances (i.e., without network connectivity) in an easy and efficient manner. This may be of particular value for large appliances such as, for example, a washer, dryer, air conditioner, dishwasher, or the like, where the costs and complexity associated with replacing the non-smart appliance with a smart appliance may be high. In some embodiments, each of the remotely-controllable switches may be controlled by a single controller (e.g., controller **2330**, described below), which may simplify installation and use of the system, and may allow a homeowner to control many electrical devices in their home using a single, convenient interface.

[0153] As shown in FIG. **23**, for example, energy control system **2310** may include one or more electrical switches **2340**. Electrical switch **2340** may be, for example, a device for interrupting the flow of current in an electrical circuit. In some embodiments, electrical switch **2340** may be or may include, for example, an overcurrent protection device, circuit breaker, fuse, electromechanical circuit breaker, electrical relay, electromechanical relay, reed relay, transistor, solid-state relay, solid-state contactor, and/or solid-state power switch.

[0154] In some embodiments, each of electrical switches **2340** may be operated independently of other electrical switches **2340** in electrical system **2300**. As shown in FIG. **23**, for example, each of electrical switches **2340** may be electrically disposed between an electrical circuit **2376** and the source of power for the electrical circuit **2376** (e.g., grid power **2380**, energy storage system **2350**, and/or power generation system **2360**). Thus, the flow of current to each of electrical circuits **2376** from grid power **2380**, energy storage system **2350**, and/or power generation system **2360**, for example, may be controlled, individually, by opening and closing the corresponding switch for electrical circuits **2376**. In some embodiments, electrical switches **2340** may be electrically disposed between loads **2370** and the source of power for loads **2370** (e.g., grid power **2380**, energy storage system **2350**, and/or power generation system **2360**). Thus, opening electrical switches **2340** may interrupt the flow of current to loads **2370** from grid power **2380**, energy storage system **2350**, and/or power generation system **2360**, for example.

[0155] In some embodiments, electrical switch **2340** may be a remotely-controllable switch **2342**. Remotely-controllable switch **2342** may be configured to be remotely controlled, which is to say that remotely-controllable switch **2342** may be operated without direct physical interaction from a user. In some embodiments, remotely-controllable switch **2342** may be or may include, for example, an overcurrent protection device, circuit breaker, electromechanical circuit breaker, electrical relay, electromechanical relay, reed relay, solid-state relay, solid-state contactor, and/or solid-state power switch that is capable of being operated remotely. In some embodiments, remotely-controllable switch **2342** may be a motorized overcurrent protection device such as, for example, a motorized circuit breaker. In some embodiments, remotely-controllable switch **2342** may be a smart switch, smart breaker, smart relay, or the like.
[0156] Energy control system **2310** may include one, two, three, four or more remotely-controllable switches **2342**. In some embodiments, each electrical switch **2340** that is disposed between a load **2370** and the source of power for the load **2370** (e.g., grid power **2380**, energy storage system **2350**, and/or power generation system **2360**) may be a remotely-controllable switch **2342**. In some embodiments, energy control system **2310** may include remotely-controllable switches **2342** as well as switches that are not remotely-controllable. In some embodiments, each electrical switch **2340** that is disposed between a non-backup load **2374** and the source of power for the non-backup loads **2374** (e.g., grid power **2380**) may not be remotely controllable. In some embodiments, each electrical switch **2340** that is disposed between a backup load **2372** and the source of power for the backup load **2372** (e.g., grid power **2380**, energy storage system **2350**, and/or power generation system **2360**) may be a remotely-controllable switch **2342**.
[0157] In some embodiments, remotely-controllable switch **2342** may be configured to be controlled, for example, by a controller **2330**. Controller **2330** may be or may include, for example, a computer, microcontroller, or other processing device. In some embodiments, controller **2330** may be a photovoltaic monitoring system (e.g., photovoltaic monitoring system **330**, described above). Remotely-controllable switches **2342** may be connected to controller **2330** using one or more control cables **2344**. In some embodiments, control cable **2344** may be, for example, a controller area network (CAN) bus, a power line communication cable (PLC), an RS485 cable, or other cable capable of sending electronic data between controller **2330** and remotely-controlled switches **2342**. In some embodiments, controller **2330** and remotely-controllable switches **2342** may be configured to communicate wirelessly and may operate on a variety of frequencies, such as Very High Frequency (e.g., between 30 MHz and 300 MHz) or Ultra High Frequency (e.g., between 300 MHz and 3 GHz) ranges, and may be compatible with certain network standards such as cell phone, WIFI™, or BLUETOOTH® wireless networks, for example.
[0158] In some embodiments, remotely-controllable switch **2342** may be configured to receive electronic data from controller **2330**, and may be configured to change state (e.g., opening or closing the switch) based on the electronic data received. In some embodiments, remotely-controllable switch **2342** may also be configured to send electronic data to controller **2330**. For example, in some embodiments, remotely-controllable switch **2342** may include a load meter (such as, e.g., load meter **340** described above) configured to monitor the flow of current through the switch (and, accordingly, through the corresponding circuit **2376**), and remotely-controllable switch **2342** may be configured to send electronic data (e.g., computer-processable data and/or information represented by an analog or digital signal) relating to the flow of current through the switch to controller **2330**.
[0159] As shown in FIG. **23**, for example, controller **2330** may include a transceiver **2332** that is configured to send and receive information wirelessly. Transceiver **2332** may allow controller **2330** to connect to a network **2390**, which may include, for example, a Wireless Local Area Network ("WLAN"), Campus Area Network ("CAN"), Metropolitan Area Network ("MAN"), or Wide Area Network ("WAN"). Transceiver **2332** may be configured to operate on a variety of frequencies, such as Very High Frequency (e.g., between 30 MHz and 300 MHz) or Ultra High Frequency (e.g.,

between 300 MHz and 3 GHz) ranges, and may be compatible with specific network standards such as cell phone, WIFI™, or BLUETOOTH® wireless networks, for example. In some embodiments, controller **2330** may connect to network **2390** using a wired connection (e.g., Ethernet or the like).

[0160] In some embodiments, controller **2330** may also be configured to connect to, control, and/or receive information from network-connected smart devices and/or smart appliances (e.g., smart outlets, smart plugs, smart bulbs, smart washers, smart refrigerators). For example, controller **2330** may send electronic data to and receive electronic data from network-connected smart devices and/or smart appliances over network **2390**. In this manner, controller **2330** may control devices and/or appliances individually, rather than controlling the entire circuit **2376** to which the devices and/or appliances are connected. Such flexibility in control may allow homeowners to tailor energy control system **2310** to fit the unique needs of their home. Further, such flexibility may allow homeowners to integrate smart devices and/or smart appliances into a single, centralized control system, which may simplify setup and use of energy control system **2310**.

[0161] In some embodiments, controller **2330** may communicate with a user device **2392** over network **2390**. User device **2392** may be, for example, a cell phone, smartphone, tablet computer, laptop computer, desktop computer, personal computer, wearable computer, smartwatch, or other computing device capable of connecting to network **2390** through a wired or wireless connection. In some embodiments, user device **2392** may connect directly to controller **2330** (e.g., using a wired or a peer-to-peer network connection).

[0162] In some embodiments, user device **2392** may include a user interface **2394**. In some embodiments, user interface **2394** includes a touch screen display for receiving user input and communicating information to the user. In some embodiments, user interface **2394** includes electromechanical buttons for receiving input from a user. In some embodiments, user interface **2394** includes a visual display for communicating with or displaying information to a user. In some embodiments, user interface **2394** includes a combination of touch screens, electromechanical buttons, and/or visual displays.

[0163] User interface **2394** may display information about, for example, the type(s) of loads **2370** in electrical system **2300**, groupings of loads **2370** in electrical system **2300**, present and/or average power consumption of loads **2370** (e.g., an energy demand associated with the plurality of loads), the state of remotely-controllable switches **2342**, the amount of power available in energy storage system **2350**, the amount of power produced by power generation system **2360**, and/or other information relating to energy control system **2310** or electrical system **2300**. User interface **2394** may receive input from the user of user device **2392** that may be used, for example, to control functions of energy control system **2310**. For example, in some embodiments, a user may control one or more remotely-controllable switches **2342** remotely via user device **2392**. In some embodiments, a user may automate remotely-controllable switches **2342** (e.g., automatically opening and/or closing at pre-set times) via user device **2392**. In some embodiments, user device **2392** may include an application configured to receive information from and send information to controller **2330**.

[0164] Controller **2330**, user device **2392**, and/or another storage device (e.g., a server connected to network **2390**) may include memory for storing information about electrical system **2300** and/or energy control system **2310**. This information may include, for example, historical data regarding the amount of power consumed by loads **2370** and the times at which the power was consumed. In some embodiments, the information may include historical data regarding the amount of power consumed by each circuit **2376** and the times at which the power was consumed. In some embodiments, the information may include, historical data regarding the amount of power consumed by individual devices and/or appliances and the times at which the power was consumed. In some embodiments, controller **2330** may determine the type of appliance, for example, based on the power profile observed by controller **2330**. In some embodiments, the information may include historical data regarding the state of remotely-controllable switches **2342**, the amount of power available in energy storage system **2350**, the amount of power produced by power generation system **2360**, and/or other information relating to energy control system **2310** or electrical system **2300**. As described in further detail below, the data collected by controller **2330** may aid in predicting future power consumption rates, which may be used to more efficiently provide power to backup loads **2372** during a power outage.

[0165] In order to comply with certain electrical codes and/or to prevent backup loads from drawing more power than a backup power system is able to provide, some backup power systems may be sized such that the backup power system is able to provide power to all of the backup loads simultaneously. In practice, however, the backup loads may not all draw power simultaneously, thus resulting in a backup power system that may be unnecessarily oversized, which may increase the expenses associated with installing and maintaining the backup power system. Alternatively, in order to utilize a smaller backup power system, a homeowner may be required to choose a limited selection of loads to which backup power is provided.

[0166] Energy control system **2310** may dynamically supply backup power to backup loads **2372**, which may allow a homeowner to provide power to backup loads **2372** using a more efficiently-sized backup power system (e.g., energy storage system **2350** and/or power generation system **2360**). In some embodiments, energy control system **2310** may provide power to certain backup loads **2372** sequentially—as opposed to simultaneously—so long as the power consumed by backup loads **2372** at any one time does not exceed the amount of power that the backup power system is capable of supplying. For example, controller **2330** may actively monitor power usage of backup loads **2372**, and may turn off certain circuits and/or smart devices if power consumption (e.g., load demand associated with respective backup loads **2372**) is too high. Further, controller **2330** may prevent circuits **2376** and/or smart device from being turned on if doing so would (or likely would) exceed power supply thresholds. As described in further detail below, energy control system **2310** may use a series of defaults, rules, and/or checks to make decisions as to which circuits and/or smart devices receive power during a power outage. In this manner, energy control system **2310** may provide backup power to a larger quantity of backup loads **2372** without necessitating an increase in overall size of energy storage system **2350** and/or power generation system **2360**.

[0167] In some backup power systems, the power lines that provide power to the backup loads may be hardwired (e.g., to an electrical panel). If, for example, a homeowner desires to reconfigure existing backup loads (e.g., to decrease the total backup load or regroup the loads to be supplied backup power), the homeowner may need to rewire portions of the electrical system, which may be difficult, time consuming, and expensive. However, since energy control system **2310** may dynamically supply backup power to backup loads **2372**, a homeowner may quickly and easily reconfigure backup loads **2370** (e.g., using an application on a smart phone) without having to rewire the system.

[0168] As shown in FIG. **23**, for example, in some embodiments, when electrical system **2300** does not receive electricity from grid power **2380** (e.g., during a power outage), microgrid interconnection device **2320** may be automatically opened (e.g., set in backup mode by controller **2330**). While disconnected from grid power **2380**, backup loads **2372** may continue to receive power from energy storage system **2350** and/or power generation system **2360**.

[0169] As mentioned above, in some embodiments, controller **2330** may open or close certain circuits **2376** corresponding to backup loads **2372** using remotely-controllable switches **2342**. In some embodiments, controller **2330** may also control other smart devices (e.g., smart outlets, smart plugs, smart bulbs, smart appliances, or the like) that are connected to network **2390** and that receive power from circuits **2376** corresponding to backup loads **2372**. Thus, controller **2330** may dynamically adjust the total power consumptions (e.g., energy demand) of backup loads **2372** by changing the state of remotely-controllable switches **2342** and/or the state of other network-

connected devices.
[0170] As explained above, the amount of power consumed by backup loads **2372** at any one time may not exceed the amount of power that energy storage system **2350** and/or power generation system **2360** may be able to provide at that time. As mentioned above, however, in order to more efficiently size energy storage system **2350** and/or power generation system **2360**, the maximum sum of all backup loads **2372** (e.g., if all backup loads **2372** were to draw power simultaneously) may be greater than the amount of power that energy storage system **2350** and/or power generation system **2360** may be able to provide. Accordingly, if energy control system **2310** loses grid power **2380** (e.g., during a grid power outage), certain circuits **2376** corresponding to backup loads **2372** may need to be rapidly disengaged in order to prevent energy storage system **2350** and/or power generation system **2360** from being overloaded. In some embodiments, controller **2330** may be configured to detect a power outage and rapidly open certain remotely-controllable switches **2342** in order to prevent energy storage system **2350** and/or power generation system **2360** from being overloaded. In some embodiments, controller **2330** may detect a power outage and open certain remotely-controllable switches **2342** within approximately 20-200 milliseconds of the detected outage. In some embodiments, controller **2330** may detect a power outage and open certain remotely-controllable switches **2342** within approximately 20-100 milliseconds of the detected outage.
[0171] FIG. **24** shows an exemplary block diagram illustrating aspects of a method of controlling an energy control system according to embodiments (e.g., the embodiment shown in FIG. **23**).
[0172] At step **2410**, controller **2330** may collect information relating to energy control system **2310** and/or electrical system **2300** during normal operation (e.g., not during a grid power outage). As described above, the information may include, for example, data regarding the amount of power consumed by circuits **2376** and the times at which the power was consumed. Further, the information may include data regarding the amount of power consumed by individual smart devices and/or smart appliances and the times at which the power was consumed. In some embodiments, controller **2330** may use the collected electronic data to determine a load average per circuit and/or a load average per smart device corresponding to discrete blocks of time throughout the day. Time blocks may be broken down, for example, into 1-hour blocks, 2-hour blocks, 3-hour blocks, or other time blocks, including, for example, user-designated time blocks (e.g., times when the user may be asleep, at home, or out of the house). In some embodiments, controller **2330** may use the collected data to determine an energy demand based on the amount of power consumed by circuits **2376**.
[0173] At step **2412**, controller **2330** may create a time-of-use library (e.g., a database or other structured set of data) that may define a circuit load average for each circuit **2376** and/or a smart device load average for each smart device with respect to the discrete blocks of time throughout the day. As described below, controller **2330** may use this information to determine which backup loads **2372** receive power as a default during a grid power outage.
[0174] At step **2414**, controller **2330** may detect a power outage. Then, in some embodiments, controller **2330** may determine a default set of circuits **2376** (and/or default smart devices disposed on circuits **2376**) to which backup power will be provided. Controller **2330** may, for example, determine the amount of power that energy storage system **2350** and/or power generation system **2360** are able to provide, the amount of power currently stored in energy storage system **2350**, and/or the amount of power currently being produced by power generation system **2360**. Controller **2330** may also query the time-of-use library described above at step **2412** to determine, for example, which circuits and/or devices will most likely require power for the instant block of time, and the amount of power that such circuits and/or devices will likely require. Based on this information, controller **2330** may then determine which circuits **2376** (and/or smart devices disposed on circuits **2376**) will receive power as a default after the power outage is detected. Then, controller **2330** may change the state of certain remotely-controllable switches **2342** and/or smart

devices in accordance with the determined default setting. In some embodiments, a user may modify the default circuits **2376** (and/or smart devices disposed on circuits **2376**) as determined by controller **2330**. In some embodiments, a user may manually choose which circuits **2376** (and/or smart devices disposed on circuits **2376**) will receive backup power as a default after a power outage is detected.

[0175] At step **2416**, after the default circuits **2376** (and/or smart devices disposed on circuits **2376**) have been determined and engaged, a user may request that certain circuits or devices other than the predetermined defaults be turned on. The user may make such a selection using, for example, user device **2392**. As described below, controller **2330** may then perform a series of calculations and/or checks to determine if the request may be fulfilled.

[0176] At step **2418**, after controller **2330** receives a selection from a user (e.g., via user device **2392**), controller **2330** may perform a series of calculations and/or checks to determine if the request may be fulfilled. For example, controller **2330** may query the time-of-use library described above to determine the expected load requirement of the selected circuit and/or device. Controller **2330** may also determine the current total power usage of backup loads **2372**. Then, controller **2330** may add the expected load requirement of the selected circuit and/or device to the current total power usage of backup loads **2372**. If the expected total load with the selected circuit and/or device engaged is greater than the amount of power that energy storage system **2350** and/or power generation system **2360** is able to provide, the user's request may be denied. Controller **2330** may also determine if engaging the selected circuit and/or device would, for example, rapidly deplete energy storage system **2350**, violate any electrical codes (e.g., exceeding the maximum allowable current passing through the wires), and/or exceed the current draw limitations of energy storage system **2350** (which may, e.g., damage energy storage system **2350**). If controller **2330** determines that engaging the selected circuit and/or device would violate the predetermined rules, the user's request may be denied. If a user's request is denied, user device **2392** may prompt the user to select a different circuit or/and device (or several circuits or/and devices) to disengage such that the backup load **2372** satisfies the necessary requirements in order to supply power to the desired circuit.

[0177] At step **2420**, after making the above determinations and receiving any necessary user input, controller **2330** may then engage and/or disengage the selected circuits or/and devices in order to complete the user's request. Controller **2330** may then monitor the power consumption of backup loads **2372** to ensure that the loads do not exceed the expected amounts. If backup loads **2372** do draw more power than energy storage system **2350** and/or power generation system **2360** may provide, controller **2330** may, for example, revert to the default state or turn off power to one or more loads in order to prevent energy storage system **2350** and/or power generation system **2360** from being overloaded.

[0178] At step **2422**, once grid power **2380** is restored, controller **2330** may then close all remotely-controllable switches **2342**, thus restoring power to all circuits **2376**.

[0179] In some embodiments, some or all of the queries, calculations, and/or determinations described above as being performed by controller **2330** may be performed by a remote computing device, such as for example, a server connected to network **2390**.

[0180] FIG. **25** shows and electrical system **2500** according to some embodiments. In some embodiments, energy control system **2510** may be configured to provide metered and controllable electrical vehicle ("EV") charger integration. For example, in some embodiments, energy control system **2510** may include an EV charger system **2590**. In some embodiments, EV charger system **2590** includes an EV charger cable **2596** for connecting an electric vehicle to energy control system **2510** in order to charge and/or discharge the electric vehicle. In some embodiments, EV charger cable **2596** may include an EV charger handle **2598** that may be configured to be connected to the charging port of an electric vehicle. In some embodiments, EV charger cable **2596** may include an EV charger connector **2594** that may be configured to be received by an EV charger port **2592** of energy control system **2510**. In some embodiments, EV charger connector **2594** may be removable from EV charger port **2592**. In some embodiments, energy control system **2510** may include a cable management system (e.g., an automatic cable retractor) for storing and organizing EV charger cable **2596**. In some embodiments, EV charger system **2590** may be a wireless charging system (e.g., an induction charging system).

[0181] In some embodiments, the status and power consumption of EV charger system **2590** may be controlled and monitored using controller **2530**, which may include features of controller **2330** described above. In some embodiments, an electrical switch **2542** (e.g., an electric relay) may be disposed between EV charger port **2592** and the source of power for charging the electric vehicle (e.g., grid power **2580**, energy storage system **2550**, and/or power generation system **2560**). In some embodiments, the state (e.g., opened or closed) of electrical switch **2542** may be controlled by controller **2530**. Further, in some embodiments, a load meter **2544** may be configured to monitor the amount of power consumed by EV charger system **2590**, and controller **2530** may be configured to receive electronic data from load meter **2544** pertaining to the amount of power consumed by EV charger system **2590**. In some embodiments, an overcurrent protection device **2540** (e.g., a circuit breaker, electromechanical circuit breaker, fuse, or the like) may also be disposed between EV charger port **2592** and the source of power for charging the electric vehicle (e.g., grid power **2580**, energy storage system **2550**, and/or power generation system **2560**).

[0182] In some embodiments, switching and metering of EV charger system **2590** may be performed by a remotely-controllable switch (such as, e.g., remotely-controllable switch **2342** described above). In some embodiments, controller **2530** may be configured to determine, for example, the charge level of a connected electric vehicle. In some embodiments, controller **2530** may be configured to control the state of charging (e.g., whether or not the vehicle is being charged, the rate of vehicle charging, and/or the type of power provided to the vehicle) of a connected electric vehicle. In some embodiments, controller **2530** may send and receive over a network (such as, e.g., network **2390** described above) electronic data pertaining to the charge level and state of charging of a connected electric vehicle. In some embodiments, the charge level and state of charging of a connected electric vehicle may be monitored and/or controlled by a network-connected user device (e.g., user device **2392** described above).

[0183] In some embodiments, the energy storage system (e.g., batteries) of an electric vehicle connected to EV charger system **2590** may be used as an energy storage system for energy control system **2510**, for example, by discharging the power from the EV. In some embodiments, during a power outage, for example, energy control system **2510** may use energy stored in a connected electric vehicle to power certain backup loads (e.g., backup loads **2372** described above). Accordingly, in some embodiments, the supplemental power provided by a connected electric vehicle may increase the total amount of power that backup power system (e.g., energy storage system **2550** and/or power generation system **2560**) is able to provide.

[0184] FIG. **26** shows an electrical system **2600** according to some embodiments. In some embodiments, portions of electrical system **2600** may be configured to be rapidly de-energized, for example, in order to comply with certain electrical codes and/or to improve safety conditions for personnel (e.g., first responders) operating in the vicinity of electrical system **2600**. For example, in some embodiments, electrical system **2600** may include a rapid shutdown switch **2612** (which may include features of rapid shutdown switch **624**, described above) that is configured to rapidly de-energize portions of electrical system **2600** (e.g., portions of energy storage system **2650** and/or power generation system **2660**). For example, during an emergency (e.g., a fire), persons such as first responders (e.g., firefighters) may use rapid shutdown switch **2612** to quickly de-energize portions of energy storage system **2650** and/or power generation system **2660** in order to safely operate in the vicinity of the de-energized portions of energy storage system **2650** and/or power generation system **2660**.

[0185] In some embodiments, certain portions of electrical system **2600** may be de-energized, for example, by opening certain electrical switches or discharging energy from certain electrical components. In some embodiments, when rapid shutdown switch **2612** is engaged (e.g., pressed), energy control system **2610** may directly or indirectly receive a command (e.g., "Power Off Signal") from rapid shutdown switch **2612**, and then microgrid interconnection device ("MID" or "ATS" or "ADS") **2620** may open (set in backup mode) in order to disconnect energy control system **2610** from grid power **2680**. In some embodiments, when rapid shutdown switch **2612** is engaged, energy storage system **2650** may be disconnected from other portions of electrical system **2600**. For example, converter **2652** may directly or indirectly receive a command (e.g., "Power Off Signal") from rapid shutdown switch **2612**, and then may, for example, open a switch or otherwise disconnect energy storage system **2650** from other portions of electrical system **2600**. In some embodiments, when rapid shutdown switch **2612** is engaged, power generation system **2660** may be disconnected from other portions of electrical system **2600**. For example, converter **2662** may directly or indirectly receive a command (e.g., "Power Off Signal") from rapid shutdown switch **2612**, and then may, for example, open a switch or otherwise disconnect power generation system **2660** from other portions of electrical system **2600**. In some embodiments, when rapid shutdown switch **2612** is engaged, energy storage system **2650** and power generation system **2660** may be disconnected from each other and from other portions of electrical system **2600** (e.g., loads **2670**), and energy control system **2610** may be disconnected from grid power **2680**.

[0186] In some embodiments, during a power outage, for example, energy control system **2610** may be configured to automatically transition from an on-grid configuration to an off-grid (e.g., micro-grid) configuration. To facilitate on-grid operation, some or all of converters **2652** and/or converter **2662** may include a grid following mode. In grid following mode, converters **2652** and/or **2662** may follow the power characteristics (e.g., voltage magnitude and frequency) of grid power **2680**, for example, by controlling the current and phase angle of power distributed by the converter. To facilitate micro-grid operation, some or all of the converters **2652** and/or converter **2662** may include a grid forming mode. In grid forming mode, converters **2652** and/or **2662** may emulate the characteristics of grid power **2680** (e.g., voltage magnitude and frequency) by controlling, for example, the voltage magnitude and frequency of power distributed by the converter.

[0187] In some embodiments, energy storage system **2650** may include a master storage unit **2654** and one or more slave storage units **2656**. In some embodiments, master storage unit **2654** may send commends (e.g., "Sync Signal") to slave storage units **2656**, for example, in order to synchronize power distribution of converters **2652** of slave storage units **2656** with power distribution of converter **2652** of master storage unit **2654**.

[0188] During a power outage, converter **2652** of master storage unit **2654** may sense a disconnection from grid power **2680** (e.g., via a voltage drop on a "Grid Sensing" wire). Then, converter **2652** of master storage unit **2654** may send a signal (e.g., "ADS Signal") to energy control system **2610** to open microgrid interconnection device **2620** (set in backup mode). Then, converter **2652** of master storage unit **2654** may cease grid following mode. Then, energy control system **2610** may open microgrid interconnection device **2620** (set in backup mode) and check for a position feedback signal (e.g., "Position Feedback Signal") to confirm that the microgrid interconnection device **2620** is open (set in backup mode). Then, the position feedback signal may be relayed to converter **2652** of master storage unit **2654**. Then, converter **2652** of master storage unit **2654** may start grid forming mode. As mentioned above, in some embodiments, the power characteristics (e.g., voltage magnitude and frequency) of converter **2652** of slave storage units **2656** may be synchronized with the power characteristics of converter **2652** of master storage unit **2654**.

[0189] After a connection to grid power **2680** is restored, converter **2652** of master storage unit **2654** may sense the grid restoration (e.g., via a voltage applied on the "Grid Sensing" wire). Then, converter **2652** of master storage unit **2654** may cease grid forming mode, and may also jitter the frequency of power in order to restart a reconnection timer. Then, converter **2652** of master storage unit **2654** may send a signal (e.g., “ADS Signal”) to energy control system **2610** to close microgrid interconnection device **2620** (set in on-grid mode). Then, energy control system **2610** may close microgrid interconnection device **2620** (set in on-grid mode) and check for a position feedback signal (e.g., “Position Feedback Signal”) to confirm that the switch is closed. Then, the position feedback signal may be relayed to converter **2652** of master storage unit **2654**. Then, converter **2652** master storage unit **2654** may start phase synchronization in order to reenter grid following mode.

[0190] Some power generation systems (e.g., photovoltaic panels or wind turbines) may produce power based on environmental factors (e.g., sun or wind) that are independent of the power demands or status of other portions of the electrical system to which the power generation system is connected. Thus, at times, the amount of power produced by the power generation system may exceed the amount of power consumed by the loads of the electrical system. In some situations, the excess power may be, for example, exported to the electrical grid and/or directed to an energy storage system. However, during a grid outage, for example, the excess power may not be exported to the grid. Although the excess power may be directed to an energy storage system, the energy storage system may store only a finite amount of energy. Thus, in some cases (e.g., during a grid outage when the energy storage system is at capacity), it may be necessary and/or desirable to limit the amount of power being produced by the power generation system so as to prevent damage to the electrical system, to prevent system faults, and/or to prevent a shutdown of the system (e.g., due to excess current, voltage, and/or frequency levels or overcharging of the energy storage system).

[0191] Similarly, for some grid-connected electrical systems, it may be desirable or required to have a net zero site consumption at the point of interconnection to the grid. For example, regulations in some jurisdictions may require that a user's local power generation system does not export more power to the grid than the user's local electrical system consumes from the grid. Thus, in a manner similar to that described above with respect to an off-grid system, some grid-connected systems may need to actively control the amount of power produced by the power generation system so that power production does not exceed power consumption.

[0192] Some embodiments as described herein may provide an electrical system including an energy control system, an energy storage system, and a power generation system. The energy control system may include a controller that is configured to communicate with the energy storage system and the power generation system. The controller may also receive real-time load consumption information (e.g., from one or more load meters **340** or from one or more remotely-controllable switches **2342** described above) and/or may use collected historical load information (e.g., stored in the time-of-use library described above) to predict load consumption. As mentioned above, the desired amount of power production of the power generation system may depend on the available storage capacity (e.g., charge capacity) of the energy storage system and/or the power consumption of the loads. In some embodiments, the available storage capacity corresponds to a difference between a total storage capacity and a current state of charge of the energy storage system. Accordingly, the controller may determine the desired power production of the power generation system based on, for example, the received or estimated information regarding the storage capacity of the energy storage system and/or the power consumption of the loads. Then, the controller may enable and/or disable the power generation system (or portions thereof) to match the desired power production. In this manner, the power production of the power generation system may be tailored dynamically and in real-time to match the needs of the electrical system, which may increase the efficiency and/or reliability of the system.

[0193] As shown in FIG. **27**A, for example, electrical system **2700** may include an energy control system **2710**, an energy storage system **2750**, a power generation system **2760**, and loads **2770** (e.g., including backup loads **2772** and non-backup loads **2774**). Energy control system **2710** may also include a microgrid interconnection device **2720** and may be interconnected to grid power **2780**. In some embodiments, energy storage system **2750**, power generation system **2760**, and loads **2770** may be interconnected via a power bus **2776**. Power bus **2776** may be, for example, one or more shared power conductors. In some embodiments, power bus **2776** may be a shared AC power bus.

[0194] In some embodiments, energy storage system **2750** and/or power generation system **2760** may be configured to communicate with and/or to be controlled by a controller **2730**. Controller **2730** may be or may include, for example, a computer, microcontroller, or other processing device. In some embodiments, controller **2730** may be a photovoltaic monitoring system (e.g., photovoltaic monitoring system **330**, described above). In some embodiments, controller **2730** may be or may include features of controller **2330** described above.

[0195] Energy storage system **2750** and/or power generation system **2760** may be connected to controller **2730** using, for example, one or more control cables **2736**. In some embodiments, control cable **2736** may be, for example, a controller area network (CAN) bus, a power line communication cable (PLC), an RS485 cable, or other cable capable of sending electronic data (e.g., computer-processable data and/or information represented by an analog or digital signal) between controller **2330** and energy storage system **2750** and/or power generation system **2760**. In some embodiments, controller **2730** and energy storage system **2750** and/or power generation system **2760** may be configured to communicate wirelessly and may operate on a variety of frequencies, such as Very High Frequency (e.g., between 30 MHz and 300 MHz) or Ultra High Frequency (e.g., between 300 MHz and 3 GHZ) ranges, and may be compatible with certain network standards such as cell phone, power-line communication, WIFI™, or BLUETOOTH® wireless networks, for example. As mentioned above, in some embodiments, controller **2730** may also receive real-time load consumption information from a load monitor **2778** via a wired (e.g., control cable **2736**) or wireless connection. Load monitor **2778** may be, for example, one or more load meters **340** and/or one or more remotely-controllable switches **2342** as described above.

[0196] As shown in FIG. **27**A, energy storage system **2750** may include one or more energy storage units **2754** (e.g., a master storage **2654** and one or more slave storage units **2656** described above (see FIG. **26**)), and each energy storage unit **2754** may include one or more batteries **2758**. In some embodiments, energy storage system **2750** may include one or more converters **2752** (e.g., a bi-directional converter). In some embodiments, converter **2752** may be or may include features of converter **2652** described above. In some embodiments, each energy storage unit **2754** may include a corresponding converter **2752**.

[0197] In some embodiments, converter **2752** may be configured to send electronic data to and/or receive electronic data from controller **2730**. In some embodiments, converter **2752** may be configured to receive electronic data (e.g. commands) from controller **2730**, and may be configured to change state (e.g., absorbing power or discharging power) based on the electronic data received. In some embodiments, converter **2752** may also be configured to send electronic data to controller **2730**. For example, in some embodiments, converter **2752** may be configured to send to controller **2730** electronic data relating to the state of energy storage system **2750** (e.g., the charge level of batteries **2758**).

[0198] Power generation system **2760** may include one or more power generation arrays **2764** (e.g., a photovoltaic panel array), and each power generation array **2764** may include one or more power generation units **2768** (e.g., a photovoltaic panel). In some embodiments, power generation system **2760** may include one or more converters **2762**. In some embodiments, converter **2762** may be or may include features of converter **2662** described above. In some embodiments, converter **2762** may be, for example, a string inverter associated with multiple photovoltaic panel (as depicted in FIG. **27**A), a microinverter associated with or part of a photovoltaic panel (as depicted in FIG. **27**D), or the like. In some embodiments, each power generation array **2764** may include a corresponding converter **2762**. In some embodiments, each power generation unit **2768** may include a corresponding converter **2762** (e.g., a microinverter). In some embodiments, converter **2762** may be a part of power generation unit **2768**. In some embodiments, one, two, three, four, or more power generation units **2768** may be interconnected to a single converter **2762** (e.g., a string inverter). In some embodiments, power generation system **2760** includes one or more power optimizers (not shown) such as, for example, DC power optimizers.

[0199] In some embodiments, converter **2762** may be configured to send electronic data to and/or receive electronic data from controller **2730**. In some embodiments, converter **2762** may be configured to receive electronic data from controller **2730**, and may be configured to change state (e.g., exporting power or not exporting power, increasing or decreasing amount of power exporting) based on the electronic data received. In some embodiments, converter **2762** may also be configured to send electronic data to controller **2730**. For example, in some embodiments, converter **2762** may be configured to send to controller **2730** electronic data relating to the state of power generation system **2760** (e.g., the amount of power being exported).

[0200] As mentioned above, controller **2730** may be configured to enable and/or disable power generation system **2760** (or portions thereof) to match the desired power production amount. For example, in some embodiments, controller **2730** may be configured to send commands to converter **2762**, and converter **2762** may change the amount of power that it exports based on the received commands. As mentioned above, in some embodiments, one, two, three, four, or more power generation units **2768** (e.g., photovoltaic panels) may be interconnected to a single converter **2762** (e.g., a string inverter). Thus, controller **2730** may limit the power export of several power generation units **2768** via communication with a single converter **2762**. In some embodiments, however, each power generation unit **2768** (e.g., a photovoltaic panel) may include a corresponding converter **2762** (e.g., a microinverter) that is in communication with controller **2730**. In this manner, controller **2730** may limit the power export of each power generation unit **2768** individually, which may facilitate relatively precise control of the total power export of power generation system **2760**.

[0201] In some embodiments, controller **2730** may also control the amount of power exported by power generation system **2760** (or portions thereof) by opening and/or closing certain remotely controllable switches (e.g., remotely controllable switches **2342** (see FIG. **23**)). In this manner, controller **2730** may enable and or disable, for example, the entire power generation system **2760**, individual power generation arrays **2764**, or individual power generation units **2678**, depending on how power generation system **2760** is interconnected to energy control system **2710**.

[0202] In order to determine the amount of power that is desired to be produced (or is acceptable to be produced) by power generation system **2760**, controller **2730** may receive electronic data from other portions of electrical system **2700** and/or may make calculations to determine the desired (or required) power output of power generation system **2760**. As shown in FIGS. **27**B and **27**C, for example, controller **2730** may receive electronic data from energy storage system **2750** and/or load monitor **2778** and may use the data to determine the desired power output **2760**A of power generation system **2760**.

[0203] With reference to FIG. **27**B, in some embodiments, controller **2730** may receive electronic data **2750**B, **2770**B related to energy storage system **2750** and loads **2770**, respectively, and may use electronic data **2750**B, **2770**B to determine the desired power output **2760**A of power generation system **2760**. For example, controller **2730** may receive electronic data **2750**B from energy storage system **2750** and electronic data **2770**B from load monitor **2778**. Electronic data **2750**B may include, for example, information relating to the amount of energy currently stored in energy storage system **2750** and/or the amount of energy that energy storage system **2750** is capable of absorbing (e.g., via charging). Electronic data **2770**B may be, for example, the amount of power that loads **2770** are using and/or are expected to use (e.g., based on the time-of-use library described above). In some embodiments, the desired power output **2760**A of power generation system **2760** may be the sum of power **2750**A capable of being absorbed by energy storage system **2750** and the power **2770**A used or predicted to be used by loads **2770**. In this manner, power generation system **2760** may simultaneously provide power to loads **2770** while charging energy storage system **2750**. Controller **2730** may then send electronic data **2760**B to power generation system **2760**. Electronic data **2760**B may include, for example, data regarding the desired amount of power to be produced, or commands to enable and/or disable certain portions of power generation system **2760** to facilitate production of the desired amount of power.

[0204] With reference to FIG. **27**C, in some embodiments, controller **2730** may receive electronic data **2750**B related only to energy storage system **2750**, and may use electronic data **2750**B to determine the desired power output **2760**A of power generation system **2760**. For example, controller **2730** may receive electronic data **2750**B from energy storage system **2750**. Electronic data **2750**B may include, for example, information relating to the amount of energy currently stored in energy storage system **2750**, the amount of energy that energy storage system **2750** is capable of absorbing (e.g., via charging), and/or the amount of energy being discharged or predicted to be discharged (e.g., based on the time-of-use library described above) from energy storage system **2750**. In some embodiments, the desired power output **2760**A of power generation system **2760** may be the power **2750**A being discharged by or predicted to be discharged by energy storage system **2750**. Controller **2730** may then send electronic data **2760**B to power generation system **2760**. Electronic data **2760**B may include, for example, data regarding the desired amount of power to be produced, or commands to enable and/or disable certain portions of power generation system **2760** to facilitate production of the desired amount of power.

[0205] FIG. **27**D shows electrical system **2700** in another configuration. Electrical system **2700** as shown in FIG. **27**D may include features and/or functionality similar to or the same as electrical system **2700** described above with reference to FIG. **27**A-**27**C. FIG. **27**A, for example, shows each power generation array **2764** to have three power generation units **2768** interconnected to a single converter **2762** (e.g., a string inverter). FIG. **27**D, for example, shows each power generation unit **2768** to include a corresponding converter **2762** (e.g., a microinverter) that may be, for example, interconnected to, disposed on, or integrally formed with the respective power generation unit **2768**.

[0206] FIG. **28** shows an example block diagram illustrating aspects of a method of controlling an electrical system according to embodiments (e.g., the embodiments shown in FIGS. **27**A-**27**C and/or FIG. **27**D).

[0207] At step **2810**, controller **2730** may receive electronic data relating to energy storage system **2750**, power generation system **2760**, loads **2770**, and/or other components of energy control system **2710** or electrical system **2700**. As described above, the data may include, for example, information relating to the amount of energy currently stored in energy storage system **2750**, the amount of energy that energy storage system **2750** is capable of absorbing (e.g., via charging), the amount of power that loads **2770** are using and/or are expected to use (e.g., an energy demand), and/or the amount of energy being discharged or predicted to be discharged from energy storage system **2750**.

[0208] At step **2812**, controller **2730** may determine the desired output power **2770**A of power generation system **2760**. As described above, in some embodiments, the desired power output **2760**A of power generation system **2760** may be the sum of power **2750**A capable of being absorbed by energy storage system **2750** and the power **2770**A (e.g., an energy demand) used or predicted to be used by loads **2770**. In some embodiments, the desired power output **2760**A of power generation system **2760** may be the power **2750**A being discharged by or predicted to be discharged by energy storage system **2750**.

[0209] At step **2814**, controller **2730** may determine the expected power output of power generation system **2760**—or portions of power generation system **2760**—based on, for example, the time of day, date, predicted weather conditions, measured weather conditions, geographic location, past power generation data, or the like.

[0210] At step **2816**, after determining the expected power output of power generation system **2760** and/or portions of power generation system **2760**, controller **2730** may determine which portions of power generation system **2760** should be enabled in order to produce the desired amount of power output. For example, controller **2730** may calculate the sum of the expected power outputs of two or more portions of power generation system **2760** (e.g., one or more power generation arrays **2764** and/or one or more power generation units **2768**), and compare the calculated sum to the desired power output in order to determine if the expected power output for the selected portions is approximately equal to (or within a certain tolerance, e.g., 1%, 2%, 3%, 4%, 5%, or some other tolerance) of the desired power output.

[0211] At step **2818**, controller **2730** may send commands (e.g., via electronic data **2760**B) to power generation system **2760** to enable and/or disable certain portions of power generation system **2760**. For example, controller **2730** may command one or more converters **2762** to begin exporting power and/or command one or more converters **2762** to cease exporting power.

[0212] At step **2820**, controller **2730** may receive actual power generation information from power generation system **2760** (e.g., via electronic data **2760**B), and may compare the actual power output to the estimated power output. If the actual power output is higher than the estimated power output, controller **2730** may send additional commands to power generation system **2760** to disable portions of the system to decrease the total power output. Similarly, if the actual power output is lower than the estimated power output, controller **2730** may send additional commands to power generation system **2760** to enable portions of the system to increase the total power output. In some embodiments, a predetermined amount of storage capacity (e.g., 1% to 10% of total storage capacity, or any combination of percentage in between) may be reserved in energy storage system **2750** (e.g., at the high state of charge end of the storage capacity) so that if the actual power output is higher than the predicted power output, energy storage system **2750** may absorb the excess energy. Further, if the actual power output is lower than the predicted power output, controller **2730** may send commands to energy storage system **2750** (e.g., via electronic data **2750**B) to discharge power to compensate for the lower than expected power output.

[0213] In some embodiments, energy storage system **2750** may include features or follow certain protocols in order to increase the efficiency of the system, to maintain operation of the system, and/or to prolong the life of the system.

[0214] With reference to FIG. **29**, for example, energy storage system **2750** may change state or operations based on the amount of energy stored in energy storage system **2750** relative to the capacity of the system (e.g., the charge level or state of charge (SOC)). As shown, in some embodiments, energy storage system **2750** may include an end-of-charge (EOC) or upper limit SOC threshold **2902** (e.g., approximately 100% charge level) where energy storage system **2750** has reached its maximum storage capacity. In some embodiments, energy storage system **2750** may include a high-end-protection threshold **2904** (e.g., any percentage of charge level within approximately 90% to approximately 99% charge level, such as approximately 99% charge level) where energy storage system **2750** may, for example, cease charging and/or take other precautions in order to prevent overcharging of the system and/or to leave a sufficient buffer to absorb sudden increases in power from power generation system **2760**. In some embodiments, energy storage system **2750** may include a near-full threshold **2906** (e.g., any percentage of charge level within approximately 90% to 99% charge level, such as approximately 97% charge level) where energy storage system **2750** may, for example, slow charging or take other precautions (e.g., disconnecting from power bus **2776**) in order to prevent overcharging of the system. In some embodiments, energy storage system **2750** may include a recharging initiation threshold **2908** (e.g., any percentage of charge level within approximately 85% to approximately 95% charge level, such as approximately 90% charge level) where energy storage system **2750** may, for example, send electronic data (e.g., electronic data **2750**B) to controller **2730** to indicate that energy storage system **2750** may be recharged. These threshold levels may be modified as appropriate for a particular energy storage system.

[0215] In some embodiments, energy storage system **2750** may include a power bus down threshold **2910** (e.g., any percentage of charge level within approximately 1% to approximately 15% charge level, such as approximately 10% charge level) where energy storage system **2750** may, for example, cease to provide power (e.g., AC power) to loads **2770** via power bus **2776** in order to conserve power. In some embodiments, energy storage system **2750** may include a battery sleep threshold **2912** (e.g., any percentage of charge level within approximately 1% to approximately 10% charge level, such as approximately 3% charge level) where energy storage system **2750** may, for example, enter a sleep mode to further conserve energy until further charging is expected or available. In some embodiments, energy storage system **2750** may be brought out of sleep mode, for example, via a wake signal (e.g., a voltage signal or a logic signal) from energy control system **2710**, which may initiate an immediate restart of energy storage system **2750** or a time-delayed restart of the system. In some embodiments, energy storage system **2750** may include an low-end-protection threshold **2914** (e.g., approximately 1% to approximately 10% charge level, such as approximately 1% charge level, or any combination of percentage in between) where energy storage system **2750** may, for example, take precautions in order to prevent damage to the system from excessive discharge. These threshold levels may be modified as appropriate for a particular energy storage system.

[0216] In some embodiments, energy storage system **2750** may include a power bus up region **2920** (e.g., between approximately 97%-10% charge level, such as approximately 95% to approximately 10% charge level, approximately 90% to approximately 15% charge level, or any combination of percentage in between) where energy storage system **2750** may, for example, provide power to loads **2770** via power bus **2776**. In some embodiments, energy storage system **2750** may include a communication up region **2930** (e.g., between approximately 10%-3% charge level, or any combination of percentage in between) where energy storage system **2750** may, for example, provide power to only the communication systems (e.g., controller **2730**) of electrical system **2700**. These threshold levels may be modified as appropriate for a particular energy storage system.

[0217] In some embodiments, the thresholds and regions described above may vary, and may be dynamically adjusted (e.g., by a user or according to program logic) based on, for example, environmental conditions, state of grid connection, battery performance, operational mode, or other factors.

[0218] In some embodiments, energy storage system **2750** may include a self-consumption mode (e.g., a solar self-consumption mode), where energy storage system **2750** may attempt to minimize usage of grid power **2780** by, for example, maximizing the amount of energy captured by power generation system **2760** and discharge from energy storage system **2750**. In self-consumption mode, energy storage system **2750** may permit a relatively wide depth of charge and discharge of batteries **2758** in order to increase the amount of energy that may be captured and utilized as a result of power generation system **2760**. In some embodiments, for example, the range of charge and discharge of batteries **2758** may be between approximately 99%-1% or any combination of percentage in between. In self-consumption mode, if energy storage system **2750** has reached its desired storage capacity (e.g., the near-full threshold), excess power produced by power generation system **2760** may be, for example, exported to electrical grid **2780** or may be curtailed (e.g., using the power limiting procedures described above) to match the demand of loads **2770** and/or the discharge of energy from energy storage system **2750**.

[0219] In some embodiments, energy storage system **2750** may include a backup mode, where energy storage system **2750** may reserve a set amount energy in energy storage system **2750** in order to maintain certain operations of electrical system **2700** during, for example, a grid power outage. In backup mode, energy control system **2710** may permit a narrower depth of charge and discharge of batteries **2758** in order to maintain prolonged operation of the system during an outage. In some embodiments, the range of charge and discharge of batteries **2758** may be between approximately 95%-10% or any combination of percentage in between.

[0220] In backup mode, energy storage system **2750** may reserve energy (e.g., communication up region) to maintain power to communication systems of electrical system **2700**. In some embodiments, such a reserve of power may be used to maintain logic communication between power generation system **2760** and energy storage system **2750**. In this manner, power generation system **2760** may detect (e.g., automatically) the sunrise and communicate the event to energy storage system **2750**. Then, energy storage system **2750** may, for example, change state or operation based on the anticipated power production of power generation system **2760**. In some embodiments, energy storage system **2750** may begin to provide power to loads **2770** via power bus **2776** even if energy storage system **2750** is below the power bus down threshold, based on a detected sunrise or based on an estimated time of sunrise. In some embodiments, energy storage system **2750** may attempt to provide power to loads **2770** via power bus **2776** based on the estimated sunrise time and, if power production is less than expected (or none), may cease to provide power to power bus **2776**. Then, energy storage system **2750** may periodically reattempt to provide power to power bus **2776** while monitoring power production. The length of each periodic reattempt to provide power may be inversely proportional to the load consumption during the reattempt (e.g., higher loads may result in a shorter attempt duration, and lighter loads may result in a longer attempt duration).

[0221] In some embodiments, energy storage system **2750** may include a time-of-use mode. Time of use mode may permit customers and utilities to support grid resilience by, for example, charging energy storage system **2750** at off-peak power use hours and discharging energy storage system **2750** at peak power use hours. In some embodiments, the depth of charge and discharge of batteries **2758** may be relatively narrow in order to increase lifespan of energy storage system **2750** due to more frequent cycling. In some embodiments, the range of charge and discharge of batteries **2758** may be between approximately 85%-20% or any combination of percentage in between.

[0222] In some embodiments, energy storage system **2750** may attempt to limit the degradation of battery sub units **2754** equally in system **2700**. For example, controller **2730** may send equal power commands to each storage converter **2752**. As another example, over the battery lifetime, battery health information received by the controller may also be used to modify the cycling characteristics across the multiple battery sub units **2758** in energy storage system **2750**.

[0223] In some embodiments, controller **2730** may command energy storage system **2750** (e.g. via converter **2752**) to increase or decrease battery charge or discharge rates for individual batteries **2758**, which may be forecasted dynamically based on forecasted PV generation and the battery recommended peak SOE level (e.g. <80%). For example, the charge rate of the battery may be reduced from a C/2 rate to a C/3 rate from forecasted photovoltaic energy generation by generation system **2760**.

[0224] Referring to FIGS. **31**A-**31**B, for example, the amount of energy stored in energy storage system **2750** may change over a period of time (e.g., a day) and may, for example, correspond to the quantity and duration of sunlight available to power generation system **2760**, the power required by loads **2770**, and/or the mode of operation of the energy storage system (e.g., self-consumption, backup, or time-of-use). FIG. **31**A, for example, shows the state of charging of energy storage system **2750** operating in a self-consumption mode (described above). FIG. **31**B, for example, shows the state of charging of energy storage system **2750** operating in a backup mode (described above). As shown, the amount of energy stored in energy storage system **2750** may increase after sunrise until energy storage system **2750** reaches its capacity (or a predetermined level). Then, power may be, for example, exported to grid **2780** and/or power production of power generation system **2760** may be curtailed as described above. After sunset, energy may be discharged from energy storage system **2750**, for example, to power loads **2770**. As shown in FIG. **31**B, for example, energy storage system **2750** may include a dynamic reserve setting, which may reserve a varying amount of energy in energy storage system **2750** (e.g., to maintain certain operations of electrical system **2700**) that may be determined and set based on, for example, the time of day at which a power outage occurs, the predicted amount of power required by loads **2770**,

or other factors.
[0225] As shown in FIG. **27**A and FIG. **30**, for example, electrical system **2700** may include a user interface **2790**. In some embodiments, user interface **2790** may include, for example, a printed circuit board (PCB) **2791**, a carrier portion **2792** (e.g., made of plastic, metal, or the like) within which the PCB **2791** may be seated and which may include adhesive to affix to electrical system **2700**, a lens **2793** (e.g., made of polycarbonate, glass, plastic, or the like) covering the PCB **2791** and/or carrier portion **2792**, and/or a silkscreen **2796** that may be attached to a front or rear side of the lens **2793** and that may include one or more symbols **2794** to provide information to a user. In some embodiments, user interface **2790** may include a visual display for communicating with or displaying information to a user. In some embodiments, user interface **2790** may include one, two, three, four, or more lights **2795** (e.g., LEDs, which may be disposed on PCB **2791**) for conveying information to a user. In some embodiments, user interface **2790** may include light pipes that receive and transfer light from lights **2795** to provide even illumination at lens **2793**. In some embodiments, the number of illuminated lights may indicate the storage level of energy storage system **2750** (e.g., 0% to 100%). In some embodiments, the color of lights may indicate the status (e.g., charging or discharging) of energy storage system **2750**. For example, green may indicate charging and yellow or red may indicate discharging. Specific lights may indicate certain status whether illuminated or non-illuminated, for example, back-up mode operation if illuminated or grid connected mode if non-illuminated. In some embodiments, the lights may turn on and off (e.g., flash) in sequence, and the rate (e.g., every 0.2 s, 0.5 s, 1 s, 2 s, or 3 s) at which the lights turn on and off may indicate, for example, the rate of charge and/or discharge (e.g., 5 KW, 3 KW, or 1 kW) of energy storage system **2750**. In some embodiments, lights **2795** may change color based on ambient conditions. In some embodiments, user interface **2790** may include a motion sensor (e.g., passive infrared sensor) to detect user motion proximate to energy storage system housing. In some embodiments, lights **2795** may change color when the motion sensor detects user motion.
[0226] In some embodiments, user interface **2790** may include electromechanical buttons for receiving input from a user. In some embodiments, user interface **2790** may include a touch screen display for receiving user input and communicating information to the user. In some embodiments, user interface **2790** may include a combination of touch screens, electromechanical buttons, and/or visual displays. In some embodiments, user interface **2790** may include motion detectors configured to detect the presence of a user. In some embodiments, user interface **2790** may include sensor configured to detect user gestures. In some embodiments, user interface **2790** may include a light sensor that detects the intensity of ambient light, and the brightness of user interface **2790**, for example, may adjust based on the intensity of ambient light. In some embodiments, a user may manipulate user interface **2790** to change the state of, or operations of electrical system **2700**.
[0227] In some embodiments, user interface **2790** may be disposed on energy control system **2710**. In some embodiments, user interface **2790** may be disposed on energy storage system **2750**. In some embodiments, electrical system **2700** may include multiple user interfaces **2790** disposed at various locations of electrical system **2700**.
[0228] In some embodiments, user interface **2790** may display information regarding the status of electrical system **2700**, or portions of electrical system **2700**. For example, in some embodiments, user interface **2790** may display information regarding the amount of energy stored in energy storage system **2750**, the mode of energy storage system **2750** (e.g., self-consumption, backup, or time-of-use), whether energy control system **2710** is connected to grid power **2780**, whether energy control system **2710** is connected to a connected to a local or remote network (e.g., network **890** described above) or server, and/or whether electrical system **2700** is operating as expected (e.g., whether or not there are any system warnings and/or alerts).
[0229] FIG. **32** shows a diagram illustrating the timing of certain operations and changes of state of certain portions of an electrical system when switching from a non-backup mode (e.g., receiving power from the electrical grid) to a backup mode (e.g., not receiving power from the electrical

grid). In some embodiments, energy control units as described herein (e.g., energy control unit **2610** described above) may permit a rapid transition from grid power to backup power (e.g., stored in energy storage system **2750** described above), and may coordinate the transition, for example, by synchronizing changing of the state of the microgrid interconnection device (e.g., microgrid interconnection device **2620** described above) with other changes to the electrical system. In some embodiments, energy control units as described herein may enable a transition from grid power to backup power in less than 1 second. In some embodiments, energy control units as described herein may enable a transition from grid power to backup power in less than 0.5 seconds. In some embodiments, energy control units as described herein may enable a transition from grid power to backup power in approximately 1/30th of a second.
[0230] FIG. **35** shows and electrical system **3500** according to some embodiments. Electrical system **3300** may include the same or similar features as the electrical system embodiments described herein. In some embodiments, electrical system **3500** may include an energy control system **3510**, an energy storage system (e.g., any one of the energy storage system embodiments shown in FIGS. **1-8** and **26-27**D) electrically coupled to energy control system **3510**, and a PV system **3360** (a power generation system) electrically coupled to energy control system **3510**. In some embodiments, energy control system **3510** may be configured to receive electrical energy from a utility grid, the energy storage system, and PV system **3560**. In some embodiments, energy control system **3310** may control the distribution of electrical energy received from energy storage system **3350**, PV system **3360**, and utility grid **384** to a plurality of loads via a backup load interconnection and a non-backup load interconnection (e.g., any one of the energy storage system embodiments shown in FIGS. **1-8** and **26-27**D).
[0231] In some embodiments, energy control system **3510** may be configured to provide metered and controllable EV charger integration. For example, in some embodiments, energy control system **3510** may include an EV charger system **3590** configured to charge and/or discharge electrical energy between energy control system **3510** and an electric vehicle **3502**. In some embodiments, EV charger system **3590** may include the same or similar features as EV charger system **2590** shown in FIG. **25**. For example, EV charger system **3590** may include an EV charger port and an EV charger cable having an EV charger connector connected to the EV charger port and an EV charger handle **2598** connected to electric vehicle **3502**.
[0232] In some embodiments, energy control system **3510** may include a controller **3530** to monitor and control the status and power consumption of EV charger system **3590**. In some embodiment, energy control system **3510** may include a PV monitoring system **3532** configured to monitor the status and performance of PV system **3560** and/or control operation of PV system **3560**. In some embodiments, controller **3530** may be linked to PV monitoring system **3532** such that controller **3530** receives electronic data related to PV system **3560**. In some embodiments, electronic data related to PV system **3560** may indicate current power output of PV system **3560** and a predicted power output of PV system **3560**.
[0233] As shown in FIGS. **36**A-C, for example, electronic data related to PV system **3560** may indicate the PV power output over the course of a selected time period (e.g., a day). As shown in FIG. **36**A for example, the power output of PV system **3560** during a typical day increases to a peak power output around midday and then decreases to approximately zero at night. However, during the same time, the home load profile (e.g., energy demand by the home) fluctuates throughout day, not necessarily matching the power output of PV system **3560** such that there is an excess between the power output of PV system **3560** and the energy demand of the home throughout the day.
[0234] In some embodiments, controller **3530** may control status and power consumption of EV charger system **3590** based on electronic data related to PV system **3560** such as adjusting the power consumption based on the available excess power output between the PV system **3560** and the energy demand of the home shown in FIG. **36**A. For example, controller **3530** may set EV

charger system **3590** to an on-demand mode or a dynamic mode. In some embodiments, under the dynamic mode, controller **3530** may prioritize the EV charger system **3590** as a special load to receive any excess power from PV system **3560** over other non-prioritized loads.

[0235] In some embodiments, controller **3530** may set in the EV charger system **3590** to the on-demand mode so that electric vehicle **3502** is charged at a predetermined rate for a period of time selected by a user. In some embodiments, when set in on-demand mode, EV charger system **3590** may allow electric vehicle **3502** to receive power from either the utility grid, the electronic storage system, or PV system **3560** to meet the user demand. As shown in FIG. **36**B for example, in some embodiments, under on-demand mode, EV charger system **3590** may charge electric vehicle **3502** at a fixed charging rate (e.g., a maximum charging rate) over a user-selected period of time, even when the monitored PV power output is minimal.

[0236] In some embodiments, controller **3530** may set the EV charger system **3590** to a dynamic mode such that EV charger system **3590** dynamically charges the electric vehicle **3502** based on the electronic data related to PV system **3560**. As shown in FIG. **36**C for example, in some embodiments, under the priority mode, EV charger system **3590** may charge electric vehicle **3502** at a dynamic rate that corresponds to the monitored PV power output of PV system **3560**. For example, as the PV power output of PV system **3560** increases, the charging rate by EV charger system **3590** increases proportionally. In some embodiments, under the dynamic mode, controller **3530** may not allow power received from the utility grid to be transferred to the EV charger system **3590**.

[0237] FIG. **37** shows an example block diagram illustrating aspects of a method **3700** of controlling EV charger system **3590** (e.g., as shown in FIGS. **25** and **35**).

[0238] In some embodiments, method **3700** may include a step of **3710** of determining whether EV charger system **3590** is operating under the on-demand mode or the dynamic mode. In some embodiment, step **3710** may include monitoring a user selection from a user device indicating whether EV charger system **3590** is operating under the on-demand mode or the dynamic mode, and the determination of operating mode may be based on the user selection. In some embodiments, step **3710** may include receiving electronic data related to PV system **3560**, the energy storage system, utility grid, and/or the plurality of loads, and the determination of operating mode may be based on the electronic data.

[0239] In some embodiments, after determining that EV charger system **3590** is set in the on-demand mode, method **3700** may include a step **3720** of charging the battery of electric vehicle **3502** at a fixed rate for a time period demanded by a user. For example, in some embodiments, a user may demand to charge the battery of electric vehicle **3502** at a maximum charge rate for a selected period of time, and in response, EV charger system **3590** charges the battery of electric vehicle **3502** at the maximum charge rate for the selected period of time. In some embodiments, step **3720** may include using EV charger system **3590** to charge the battery of electric vehicle **3502**.

[0240] In some embodiments, after determining that EV charger system **3590** is set in the dynamic mode, method **3700** include a step **3730** of receiving electronic data related to PV system **3560** and/or the plurality of loads. In some embodiments, the electronic data related to the PV system **3560** may include electronic data captured by PV monitoring system **3532**. In some embodiments, the electronic data related to the plurality of loads may be data captured by load meters or data stored in memory of controller **3530**.

[0241] In some embodiments, method **3700** may include a step **3740** of determining current PV power output of PV system **3560** and a load demand based on the electronic data received in step **3730**. In some embodiments, step **3740** may include applying one or more algorithms to the electronic data received in step **3730** to determine the load demand and the current PV power output.

[0242] In some embodiments, method **3700** may include a step **3750** of determining whether the current PV power output is greater than the load demand. In some embodiments, the current PV power output and the load demand used in step **3750** are based on the determinations made in step **3740**.

[0243] In some embodiments, if the current PV power output is greater than the load demand, method **3700** may include a step **3760** of charging the battery of electric vehicle **3502** at a dynamic rate corresponding to the excess difference between the current PV power output and the load demand. For example, in some embodiments, the charging rate under step **3760** may be proportional to the excess difference, such that as the difference between the current PV power output and the demand increases, the charging rate increases, as well. In some embodiments, step **3760** may include using the EV charge system **3590** to charge the battery of electric vehicle **3502**. In some embodiments, step **3750** may include interrupting power distribution from the utility grid to the EV charger system **3590**.

[0244] In some embodiments, if the current PV power output is less than the load demand, method **3700** may include a step **3770** of disabling charge to the battery of electric vehicle **3502**. In some embodiments, step **3770** may include disabling the EV charger system **3590** to disable charge to electric vehicle **3502**. In some embodiments, step **3770** may include actuating a switch electrically coupled to EV charger system **3590** to an open position.

[0245] FIG. **40** shows an electrical system **4000** according to some embodiments. Electrical system **4000** may include the same or similar features as the embodiments of the electrical systems described herein (e.g., any one of electrical systems **100-3300**). In some embodiments, electrical system **4000** may include an energy control system **4010**. In some embodiments, electrical system **4000** may include an energy storage system **4050** electrically coupled to energy control system **4010**. In some embodiments, electrical system **4000** may include a backup PV system **4060** electrically coupled to energy control system **4010**. In some embodiments, electrical system **4000** may include a plurality of loads **4070** electrically coupled to energy control system **4010**. In some embodiments, the plurality of loads **4070** may be separated into a plurality of backup loads **4072** and a plurality of non-backup loads **4074**. In some embodiments, electrical system **4000** may include a non-backup PV system **4090** electrically coupled to energy control system **4010**.

[0246] In some embodiments, backup PV system **4060** may include one or more power generation arrays **4064** (e.g., a photovoltaic panel array), and each power generation array **4064** may include one or more power generation units **4068** (e.g., a photovoltaic panel). In some embodiments, backup PV system **4060** may include one or more PV converters **4062**. In some embodiments, PV converter **4062** may include the features of any one of the converters (e.g., converters **2662**, **2762**) described herein.

[0247] In some embodiments, non-backup PV system **4090** may include one or more power generation arrays **4094** (e.g., a photovoltaic panel array), and each power generation array **4094** may include one or more power generation units **4098** (e.g., a photovoltaic panel). In some embodiments, non-backup PV system **4090** may include one or more PV converters **4092**. In some embodiments, PV converter **4092** may include the features of any one of the converters (e.g., converters **2662**, **2762**) described herein.

[0248] In some embodiments, energy storage system **4050** may include one or more storage units **4052**. In some embodiments, storage unit **4052** may include one or more batteries **4058**. In some embodiments, storage unit **4052** may include a storage converter **4054** configured to adjust a charging rate and a discharging rate of the one or more batteries **4058**.

[0249] In some embodiments, energy control system **4010** may include a backup power bus **4040** electrically coupled to backup PV system **460** via a backup PV interconnection **4011**, energy storage system **4050** via a storage interconnection **4012**, and the plurality of backup loads **4072** via a backup load interconnection **4013**. In some embodiments, energy control system **4010** may include a non-backup power bus **4080** electrically coupled to the plurality of non-backup loads **4074** via a non-backup load interconnection **4014**, non-backup PV system **490** via a non-backup PV interconnection **4015**, and a utility grid **4084** via a grid interconnection **4016**. Any one of interconnections **4011-4016** may include an AC bus, a panel, a sub-panel, a circuit breaker, any type of conductor, or a combination thereof.

[0250] In some embodiments, energy control system **4010** may include a microgrid interconnection device **4020** (e.g., an automatic transfer or disconnect switch) electrically coupled to backup power bus **4040** and non-backup power bus **4080**, such that microgrid interconnection device **4020** is electrically coupled to backup PV interconnection **4011**, storage interconnection **4012**, backup load interconnection **4013**, non-backup load interconnection **4014**, non-backup PV interconnection **4015**, and grid interconnection **4016**. In some embodiments, microgrid interconnection device **4020** may include any combination of switches, relays, and/or circuits to selectively connect and disconnect interconnections **4011-4016**.

[0251] In some embodiments, microgrid interconnection device **4020** may be configured to operate under an on-grid mode, in which microgrid interconnection device **4020** electrically connects the backup power bus **4040** to the non-backup power bus **4080**. In some embodiments, when operating under the on-grid mode, microgrid interconnection device **4020** may be configured to distribute electrical energy received from utility grid **4084** and non-backup PV system **4090** to backup loads **4072**. In some embodiments, when operating under the on-grid mode, microgrid interconnection device **4020** may be configured to distribute electrical energy received from energy storage system **4050** and backup PV system **4060** to non-backup loads **4074**.

[0252] In some embodiments, microgrid interconnection device **4020** may be configured to operate under a backup mode, in which microgrid interconnection device **4020** electrically disconnects non-backup power bus **4080** from backup power bus **4040**. In some embodiments, when operating under the backup mode, microgrid interconnection device **4020** may disrupt electrical energy received from non-backup PV system **4090** from reaching backup loads **4072**. In some embodiments, when operating under the backup mode, microgrid interconnection device **4020** may disrupt electrical communication between backup loads **4072** and utility grid **4084**. In some embodiments, when operating under the backup mode, microgrid interconnection device **4020** may disrupt electrical energy received from energy storage system **4050** and backup PV system **4060** from reaching non-backup loads **4074**.

[0253] In some embodiments, energy control system **4010** may include a controller **4022** in communication with microgrid interconnection device **4020** and configured to control the distribution of electrical energy between energy storage system **4050**, backup PV system **4060**, the plurality of loads **4070**, utility grid **4084**, and non-backup PV system **4090**. In some embodiments, controller **4022** may be configured to detect the status (e.g., power outage or voltage restoration) of grid interconnection **4016** and switch microgrid interconnection device **4020** between the on-grid mode and the backup mode based on the status of grid interconnection **4016**. If the status of grid interconnection **4016** indicates a power outage, controller **4022** may be configured to switch microgrid interconnection device **4020** to the backup mode. If the status of grid interconnection **4016** indicates a voltage restoration, controller **4022** may be configured to switch microgrid interconnection device **4020** to the on-grid mode.

[0254] In some embodiments, energy control system **4010** may include a PV monitoring system **4030** in communication with backup PV system **4060** and/or non-backup PV system **4090** such that PV monitoring system **4030** receives electronic data related to backup PV system **4060** and/or non-backup PV system **4090**. In some embodiments, controller **4022** may be linked to PV monitoring system **4030** to receive the electronic data related to backup PV system **4060** and/or non-backup PV system **4090**. In some embodiments, controller **4022** may control distribution of energy based on the electronic data related to backup PV system **4060** and/or non-backup PV system **4090**.

[0255] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present embodiments as contemplated by the inventor(s), and thus, are not intended to limit the present embodiments and the appended claims in any way.

[0256] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0257] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

Claims

1.-**6**. (canceled)

7. An energy control system, comprising: a grid interconnection electrically coupled to a utility grid; a backup power interconnection electrically coupled to a backup power source; a backup load interconnection electrically coupled to at least one backup load; a microgrid interconnection device electrically coupled to the grid interconnection, the backup power interconnection, and the backup load interconnection; and a controller in communication with the microgrid interconnection device, wherein the controller is configured to: monitor energy stored in the backup power source; cause storage of energy from the utility grid or solar according to a backup reserve value that is an amount of energy stored to maintain operations of the energy control system; and in response to detecting a grid outage, control power distribution to at least some backup loads during the grid outage to prolong the backup reserve value.

8. The energy control system of claim 7, wherein the backup reserve value is between 0% and 100% of power received by the energy control system.

9. The energy control system of claim 7, wherein the controller is further configured to provide an indication of providing power, wherein the indication includes a light emitting diode (LED).

10. The energy control system of claim 7, wherein the microgrid interconnection device is configured to switch between: an on-grid mode electrically connecting the grid interconnection and the backup power interconnection to the backup load interconnection, and a backup mode electrically disconnecting the grid interconnection from the backup power interconnection.

11. The energy control system of claim 10, wherein in response to detecting the grid outage, the controller is configured to switch the microgrid interconnection device from the on-grid mode to the backup mode.

12. The energy control system of claim 10, wherein the controller is in communication with a user device, and upon receiving an input from the user device, the controller is configured to switch the on-grid mode to the backup mode.

13. A method, comprising: under control of an energy control system comprising one or more controllers configured to execute specific instructions, monitoring energy stored in a backup power source; causing storage of energy from a utility grid or solar according to a backup reserve value that is an amount of energy stored to maintain operations of an energy control system; and in response to detecting a grid outage, controlling power distribution to at least some backup loads during the grid outage to prolong the backup reserve value.

14. The method of claim 13, wherein the backup reserve value is between 0% and 100% of power received by the energy control system.

15. The method of claim 13, further comprising providing an indication of providing power, wherein the indication includes a light emitting diode (LED).

16. The method of claim 13, further comprising instructing a microgrid interconnection device to switch between: an on-grid mode electrically connecting a grid interconnection and a backup power interconnection to a backup load interconnection, and a backup mode electrically disconnecting the grid interconnection from the backup power interconnection.

17. The method of claim 16, further comprising in response to detecting the grid outage, switching the microgrid interconnection device from the on-grid mode to the backup mode.

18. The method of claim 17, further comprising, in response to receiving an input from a user device, switching the on-grid mode to the backup mode.

19. An energy control system comprising: a grid interconnection coupled to a utility grid; a backup power interconnection coupled to at least one backup power source; a backup load interconnection coupled to at least one backup load; a non-backup load interconnection coupled to at least one non-backup load; and a microgrid interconnection device electrically coupled to the grid interconnection, the backup power interconnection, and the backup load interconnection; and a controller in communication with the microgrid interconnection device, wherein the controller is configured to: detect a power outage; and in response to detecting the power outage, instruct the microgrid interconnection device to switch between: (1) an on-grid mode that causes the grid interconnection to provide power to the at least one non-backup load by using the utility grid and causes the backup power interconnection to provide power to the at least one backup load by using a backup power source of the at least one backup power source, and (2) a backup mode that causes disconnecting of the grid interconnection and causes the backup power interconnection to provide power to the at least one backup load by using the backup power source.

20. The energy control system of claim 19, wherein the microgrid interconnection device is further configured to, in response to backup power exceeding power used by the at least one backup load, provide power to charge a battery.

21. The energy control system of claim 19, wherein the microgrid interconnection device is further configured to send a command to adjust power from a solar power inverter.

22. The energy control system of claim 21, wherein the microgrid interconnection device is further configured to send the command to indicate to the solar power inverter to reduce power provided from the solar power inverter.

23. The energy control system of claim 22, wherein the microgrid interconnection device is further configured to, in response to sending the command, cause the solar power inverter to curtail power.

24. The energy control system of claim 19, wherein the microgrid interconnection device is further configured to control a relay to isolate the backup power interconnection from the grid interconnection.

19. The energy control system of claim **19**, wherein the microgrid interconnection device is further configured to cause display of information including performance of the energy control system.

26. The energy control system of claim 19, wherein the backup power source comprises a photovoltaic (PV) power generation system, and the backup power interconnection is configured to receive electrical energy generated by a PV power generation system.